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Tellurium nanowire retinal nanoprosthesis improves vision in models of blindness

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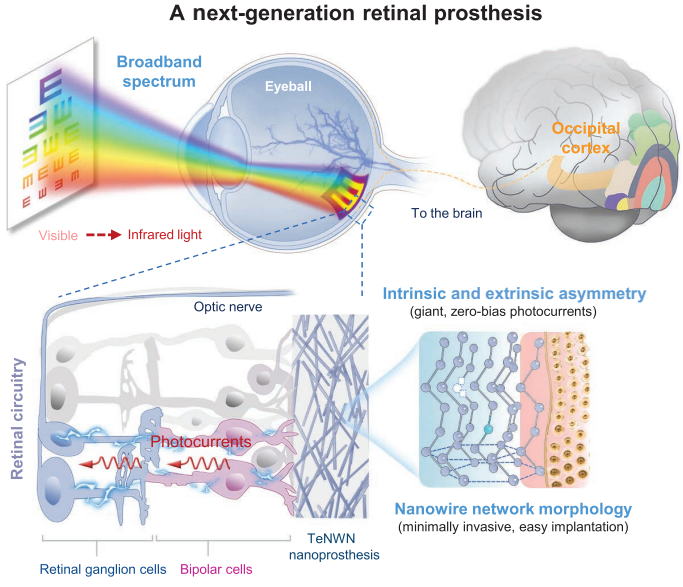
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INTRODUCTION: In nature, there are animals, such as snakes, that assess their environment more accurately by sensing both infrared radiation and the visible-light spectrum. The human eye lacks photoreceptors responsive to the infrared spectrum, and infrared light with a longer wavelength and lower energy cannot trigger visual signals. In patients with severe eye diseases (e.g., macular degeneration), infrared vision could, in principle, aid vision in low light and darkness. Developing technologies that use a wider spectrum of light, including infrared, could provide notable benefits.

RATIONALE: Present designs for broad-spectrum retinal protheses use nanoparticles or photodiodes to convert infrared into visible light or heat to stimulate retinal cells. Because they require injections or bulky auxiliary devices, there are safety and practicality issues for potential application to humans. Creation of a safe, easy-to-implant retinal prosthesis that enables the processing of both visible and infrared light may restore vision loss and enhance natural vision. We designed a next-generation retinal nanoprosthesis based on tellurium nanowire networks that intrinsically converts broadband light—including visible to infrared light—with efficient photovoltaic conversion, yielding giant with photocurrents under zero-electric bias and without the need for extra auxiliary equipment. We then achieved safe and simple implantation in the subretinal space of mice and *Macaca fascicularis*.

RESULTS: Through theoretical calculations, we show spontaneous giant and wide-spectrum photocurrents of tellurium nanowire networks to be correlated with the asymmetry of nanowire lattice internal defects and external interfacial effects. Through a combination of narrow bandgaps, strong absorption, and engineered asymmetries, tellurium optoelectronic nanodevices showed record-high photocurrents and the widest spectrum of responsive photosensitivity wavelengths compared with reported techniques for the restoration of photosensitivity in blindness, covering the visible to near-infrared–II range. Preimplantation tests confirmed the stability of the nanoprosthesis's optoelectronic properties and its precise response to light patterns. In blind mice, the implanted nanoprosthesis replaced damaged photoreceptors and triggered responses in both the optic nerve and visual cortex. Implanted mice showed better light-induced pupil reactions and improvement in light-associated learning behaviors (such as water reward–based visual-cue associative learning and choice-box tasks) when compared with untreated mice and when using light intensities nearly 80 times weaker than the clinical safety threshold. The biocompatibility and efficacy of the proposed nanoprosthesis was further demonstrated in nonhuman primates (*Macaca fascicularis*), where the nanoprosthesis was tightly bound to the retina in the subretinal space and generated robust retina-derived responses to visible and infrared light.

CONCLUSION: Our study provides biologically feasible parameters for a retinal prosthesis using designed tellurium nanowire networks.



A next-generation nanoprosthesis that restores and enhances vision. Tellurium possess broad-spectrum optical absorption from visible to infrared light (top left). A subretinal-implanted tellurium nanoprosthesis replaces degenerated photoreceptors and generates photocurrents to activate residual retinal circuitry (bottom left) and the occipital cortex (top right). Giant, spontaneous, bias-free photocurrents and minimally invasive easy implantation is achieved by asymmetry engineering and nanowire network morphology (bottom right). Together, these properties make tellurium nanowire networks (TeNWNs) the next generation of visual prosthesis technology.

These nanowires naturally convert light into photocurrent signals with zero electrical bias and can cover the visible to infrared spectrum. The tested nanoprosthesis generates strong photocurrents to activate the remaining retinal circuitry in a dysfunctional eye, works through a simple subretinal implantation procedure, and avoids bulky intra- and extraocular components. In blind mice, this retinal nanoprosthesis restored the brain's response to light and improved vision-based behaviors at clinically safe light levels. Nonhuman primates implanted with this nanoprosthesis gained infrared vision without impairment of normal vision. This successful animal study paves the way for future human trials, showcasing the potential of this prosthesis to restore visible vision and expand augmented infrared perception for blind humans and offer a safer, more effective, and wider-spectrum solution than existing technologies. □

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Tellurium nanowire retinal nanoprosthesis improves vision in models of blindness

Shuiyuan Wang[1,2]†*, Chengyong Jiang[1]†, Yiye Yu[3]†, Zhenhan Zhang[1,4]†, Ruge Quhe[5]†, Ruyi Yang[1], Yufei Tian[1], Xindong Chen[1], Wenqiang Fan[1], Yinge Niu[5], Biao Yan[1], Chunhui Jiang[1], Yang Wang[2], Zhen Wang[3], Chunsen Liu[1,2], Weida Hu[3]*, Jiayi Zhang[1]*, Peng Zhou[1,2]*

Present vision restoration technologies have substantial constraints that limit their application in the clinical setting. In this work, we fabricated a subretinal nanoprosthesis using tellurium nanowire networks (TeNWNs) that converts light of both the visible and near-infrared–II spectra into electrical signals. The broad-spectrum coverage is made possible by a combination of narrow bandgaps, strong absorption, and engineered asymmetries. Implanted into blind mice, the TeNWNs restored pupillary reflexes and enabled visually cued learning under visible and near-infrared 1550-nanometer light. In nonhuman primates, TeNWNs elicited robust retina-derived neural responses, confirming biocompatibility and feasibility. By restoring lost photosensitivity and extending vision to near-infrared, this nanoprosthesis offers a promising approach for restoring vision.

Humans cannot naturally detect infrared (IR) light because our rod and cone photoreceptors only capture higher-energy photons in the visible (VIS) range (~1.6 to 3.1 eV) (*1*). This means that longer wavelength bands with lower photon energy (such as IR) fail to induce effective bioelectric signals in photoreceptor cells (*2*). Stated another way, the human eye is not responsive to IR light.

Yet IR vision could offer advantages in the real world, including improved object detection in low-light conditions and enhanced contrast. Thus, vision restoration modalities that make greater use of the available electromagnetic spectrum, including IR bands, could provide substantial benefits (*3*).

Optoelectronic nanomaterials offer spontaneous photoresponsivity, making them ideal candidates for artificial photoreceptors. Previous work demonstrated that up-conversion nanoparticles, when bound to functional photoreceptors, can convert near-infrared (NIR) light into VIS light (*1*), whereas nanorods, coupled to temperature-sensitive channels, can absorb NIR light and convert it into heat to activate residual photoreceptors and their downstream circuitry (*4*). These techniques primarily target the augmentation of normal vision; however, intraocular injection of nanomedicine is an invasive procedure for people with normal eyesight. Hence, developing a nanomaterial with broadband-sensitivity that serves the purpose of both VIS eyesight restoration and NIR light sensitivity may be an ideal option (*5*).

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Vision restoration technologies such as stem cell replacement (*6*, *7*), targeted genetics (*8*), and optogenetics (*9*, *10*) provide partial relief but face challenges such as interference with residual retinal circuitry (*11*), integrity (*12*) and long-term effectiveness (*13*) of introduced genes, and limited therapeutic efficacy. Likewise, although immense progress has been made in the development of conventional photodiode retinal prostheses (*14*–*18*), they require redundant intraocular power supply and/or cables and extraocular goggles with integrated cameras, receivers, and controller modules (*19*), which limit real-world adaptation. Moreover, the reported wavelength and photothermal conversion of optoelectronic nanomaterials is inefficient, and the photovoltaic efficiency of photodiodes is largely restricted by the Shockley-Queisser limit (*18*). This inefficiency means that higher-energy photons are required to restore photosensitivity, which fundamentally results in a narrow bandwidth that cannot exceed the NIR-I band (≤~1000 nm). A more ideal solution would be a biocompatible implant that spontaneously generates electrical signals without external bias under both VIS and IR illumination, functioning much like healthy photoreceptors but extending photosensitivity beyond the natural range.

In this work, we created a subretinal nanoprosthesis using tellurium nanowire networks (TeNWNs) that can achieve high-efficiency photovoltaic conversion, producing a record-high photocurrent density up to 30 A cm^{-2} and exhibiting broad photosensitivity from VIS to NIR-II wavelengths up to 1550 nm. We tested this approach in mouse and nonhuman primate models of blindness.

Vision restoration and augmentation

We used optoelectronic nanomaterial tellurium (Te) to simultaneously address the three critical challenges for vision restoration and augmentation: (i) The TeNWNs form an interlaced network architecture that allows easy implantation to compensate for photoreceptor loss (tested in a retinitis pigmentosa mouse model); (ii) Te's narrow bandgap (~0.3 eV) (*20*, *21*) and high optical absorbance (up to 9% per layer) (*22*) enable responsiveness at longer wavelengths and lower photon energy, and (iii) internal (vacancy and substitution defects) and external (interfacial effects between the materials and retinal cells) asymmetries in Te drive zero-bias photocurrents, eliminating the need for any auxiliary components. Together, these properties make TeNWNs an ideal nanoprosthesis for VIS–NIR-II vision (fig. S1).

Lattice structure and photocurrent calculations of Te

A number of methods have been reported for synthesizing Te nanomaterials, including hydrothermal (*23*), thermal evaporation (*24*), and chemical or physical vapor deposition (*25*, *26*) as well as formation through solid and liquid sources (*27*, *28*). Here, we synthesized TeNWNs through chemical vapor deposition based on $SnTe_2$ powders. The crystal structure of Te, illustrated in Fig. 1A, consists of helical chains arranged in a hexagonal array, where each Te atom is covalently bonded to two neighbors along the chain and weakly bonded to atoms in neighboring chains by van der Waals interactions. Thus, Te can be viewed as a stack array of single helical chains stretching along the *c* axis in a direction perpendicular to the *b*-*c* (*a*-*c*) plane.

A scanning electron microscopy (SEM) image of the TeNWNs (Fig. 1B, left, with additional images in fig. S2) shows randomly distributed Te nanowires forming an interconnected nanonetwork. An enlarged single nanowire is shown in Fig. 1B, middle, where the *c* axis is the nanowire radial direction. Atomic force microscopy showed a typical wire thickness of ~150 nm (fig. S3), and Raman spectroscopy confirmed Te vibrational modes (fig. S4). High-resolution high-angle angular dark-field–scanning transmission electron microscopy (HAADF-STEM) of a cross section taken along the *c* axis (Fig. 1B, right), clearly resolves the expected lattice structure, in agreement with Fig. 1A. Although this noncentrosymmetric chain structure could induce a bulk photovoltaic effect, shift current along the *c* axis remains zero because the intrinsic Te has a twofold rotational axis perpendicular to the *c* axis

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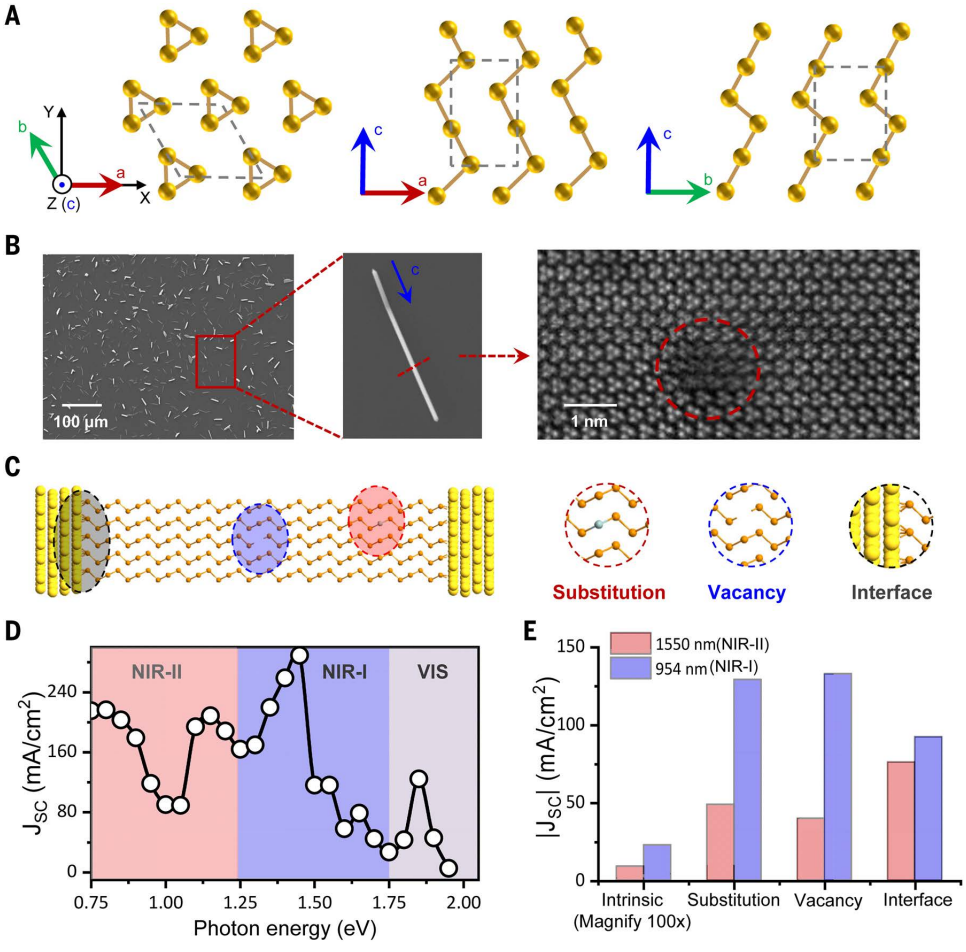


Fig. 1. Lattice and photocurrent calculations of Te. (**A**) Schematics of Te-chain asymmetric atomic structure. The gray dashed boxes indicate the unit cell of each structure. (**B**) SEM (left) and HAADF-STEM (right) for TeNWNs. The middle panel shows a magnified view of a single Te nanowire from the boxed region of the left image, radially oriented in the *c*-axis direction. The STEM image shows the defects as indicated by the red dashed circles. (**C**) The proposed device model for Te photocurrent calculations. Red, blue, and gray circles highlight the internal defects (that is, Sn substitutions, Te vacancies) and external interface effects (electrode channel), respectively. (**D**) Photocurrent response of Te under zero bias, covering VIS to NIR-II wavelengths. The simulated model for photocurrent calculations consists of Au electrodes and a 12-nm-long Te channel with a Sn substitution. J_{SC}, short-circuit current density. (**E**) Calculated photocurrent densities of intrinsic, substitution, vacancy, and interface models in NIR-I (954 nm) and NIR-II (1550 nm) bands. Sn substitution or Te vacancy defects and electrode-channel interface effects all enhance the photocurrents. The intrinsic model refers to the one with heavily doped Te as the electrodes in photocurrent calculations, and the photocurrent densities in the intrinsic model are magnified 100 times for visibility.

(*29*). Owing to the existence of stannum (Sn) precursor during the fabrication process (*30*), we found some internal defects (that is, Sn substitutions and Te vacancies) in the grown crystal samples. A HAADF-STEM image (Fig. 1B, right) shows the defects in Te (red dashed circle). We further confirmed and quantified the presence of Sn substitutions using inductively coupled plasma mass spectrometry (ICP-MS; table S1) and SEM-TEM energy-dispersive spectroscopy (EDS) mapping (fig. S5 and tables S2 and S3). It is worth noting that there are some discrepancies between the different methods: ICP-MS may be affected by incomplete dissolution of the sample, introduction of substrate impurities, and potential reactions of Sn, which can lead to discrepancies with microscopic SEM-TEM–EDS results. Owing to the asymmetric distribution of internal defects, the composition of the Te helical chain changes, which leads to broken symmetry. Such asymmetries combined with interfacial or edge effects may produce photocurrents along the *c* axis.

To understand how internal defects and external interfaces may influence the photocurrents generated, we performed quantum transport device simulations. We simulated two device models (models I and II), both of which consist of two semi-infinite electrodes and a Te channel with vacancy and substitution defects (Fig. 1C). Model I uses heavily doped Te electrodes (a homogeneous interface), whereas model II uses bulk Au electrodes (a heterogeneous interface) (fig. S6). The Te-Au interface in model II serves as a proxy for the heterogeneous contacts formed by TeNWNs and retinal cells after implantation (that is, interfacial asymmetry). Under zero bias, the device without defects (model I) exhibits only a weak photocurrent density (1 mA cm^{-2}). This weak response along the *c* axis is consistent with previous arguments that the conductivity tensor of shift current is zero (*29*). By contrast, the inclusion of defects and heterointerfaces (model II) notably boosts photocurrent under zero bias, spanning from the VIS to the NIR-II range, owing to the narrow bandgap and strong optical absorption of Te. The maximum photocurrent density reaches 288 mA cm^{-2} under illumination with a photon energy of 1.45 eV (Fig. 1D). Such enhancement of the photovoltaic responses at zero bias due to the structural disorder has also been found in other low-dimensional systems (*31*).

Formation-energy calculations (supplementary text 1 and table S4) indicate that Sn substitutions are energetically favored over Te vacancies. Yet either defect, Sn substitutions or Te vacancies, raises the zero-bias photocurrent density in the NIR from 40 to 133 mA cm^{-2}, well above that of intrinsic Te (Fig. 1E). These defects induce localized states near the Fermi level of Te, which contribute to the increased photocurrent, as indicated by the effective transmission coefficients in fig. S7. The experimentally wide-region EDS mapping and the corresponding Sn distributions show that Sn elements are enriched at the tip of the nanowire and decrease with extension (fig. S8), indicating an asymmetric distribution. We further analyzed the effect of the specific position of defects along the Te channel on the optical response theoretically: We built Te device models (both models I and II) with defects in the middle and right of the channel, respectively (figs. S9 to S11). The photocurrent densities calculated for the latter are several to a hundred times higher than that for the former, regardless of the defect type. In addition, calculations suggest that increasing defect density likewise increases the photocurrent relative to intrinsic Te (fig. S12). Overall, these results demonstrate how both interfacial asymmetry and internal defects synergistically generate robust giant photocurrents in Te.

Photocurrents in the experimental device and electrophysiology

We measured Te photocurrents using both a device optoelectronic platform and an ex vivo electrophysiological approach. For the device-based measurements, we used a metal electrode structure consistent with our calculations for model II. The output curves under NIR-II

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illuminations at varying intensities are shown in Fig. 2A. Spontaneous photocurrent densities were observed at zero bias (red dashed line in Fig. 2A) and increased with laser intensity (see fig. S13 for the current output of Te under nonillumination conditions and fig. S14 for detailed photocurrent density output curves at varying laser intensities). Furthermore, Fig. 2B shows photocurrent density curves as functions of various laser intensities in different spectral bands, including VIS, NIR-I, and NIR-II; for all tested wavelengths, photocurrent densities increased linearly with laser intensities (see fig. S15 for the dynamic responses of VIS, NIR-I, and NIR-II photocurrent densities). These photogenerated carriers are driven by the interfacial and internal electric fields without any external bias. This intrinsic ability to convert photonic energy into zero-bias electrical current is what distinguishes TeNWNs from existing photodiode retinal prostheses, as it eliminates the need for external modules.

Next, we compiled a comparative analysis of photocurrent densities generated by various materials under zero bias (Fig. 2C), including one-dimensional (1D) (*32*), 2D (*33*), and 3D (*34*–*41*) structures. The results show that dimensionality reduction markedly increases photocurrent density. Among the structures tested, TeNWNs showed the highest photocurrent density available, at nearly 30 A cm^{-2} under laser stimulation, which should ensure effective activation of residual retinal circuitry after implantation. These photocurrents are likely due to the built-in asymmetry from both interfacial effects and internal defects, as discussed in the photocurrent calculation section. As an additional control, we monitored temperature changes of the Te devices under NIR-II laser irradiation (fig. S16) and found no thermal temperature increase. We also summarized the restored responsive wavelengths in retinal prosthesis studies using different materials and mechanisms (*1*, *4*, *14*–*16*, *42*, *43*) (Fig. 2D) and showed that TeNWNs stand out among all the retinal prostheses as having the widest responsive range, from VIS into NIR-II.

For the ex vivo electrophysiological approach, we designed a testing platform (Fig. 2E) to evaluate electrical field distribution on the TeNWNs surface under various visual stimuli patterns. We projected shapes (Fig. 2F), handwritten digits, and "E" optotypes (fig. S17) using a 635-nm laser at 42.5 mW mm^{-2} onto the TeNWNs. Figure 2F and fig. S17 show the distribution of photocurrents from TeNWNs, exhibiting precise patterns.

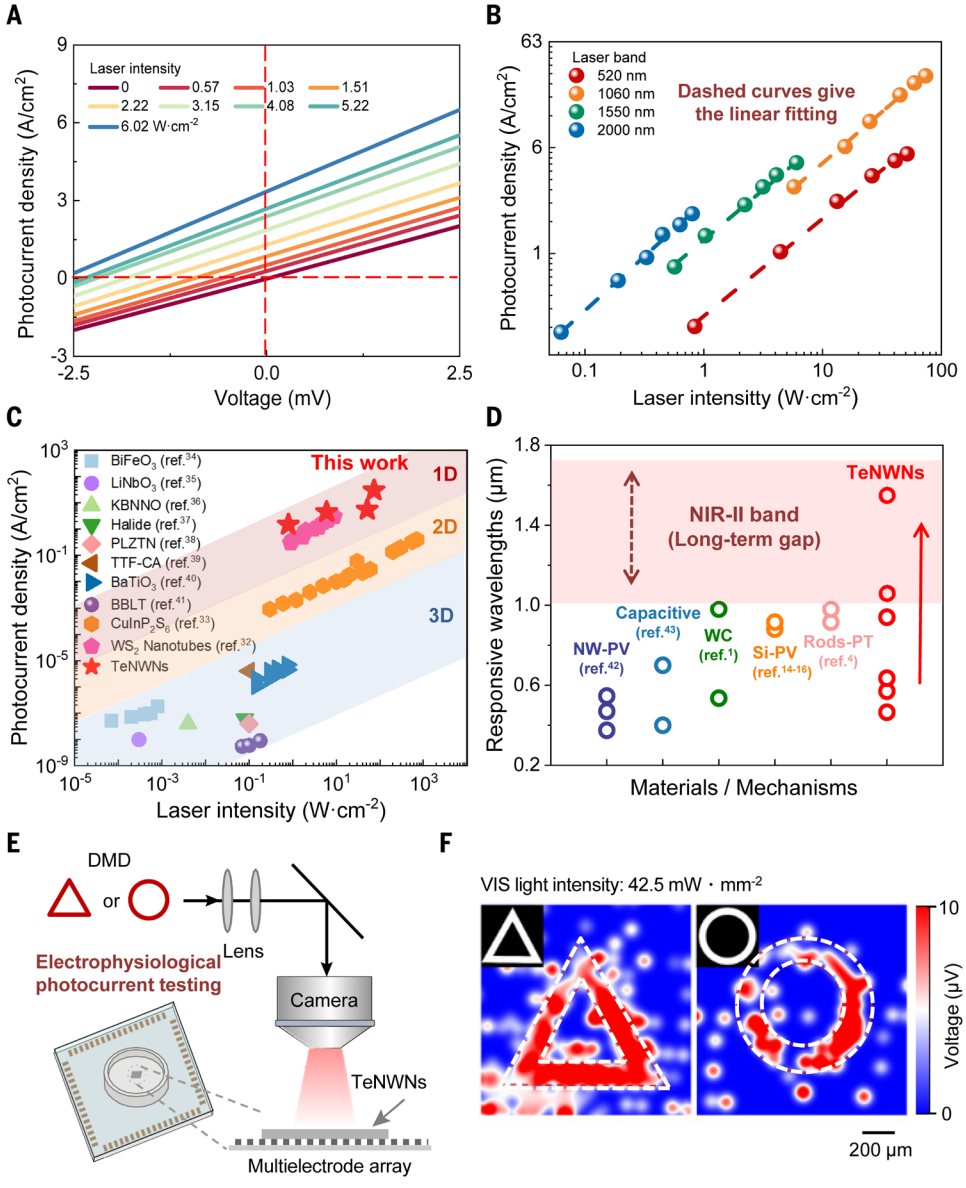


Fig. 2. Experimentally measured photocurrents under optoelectronic and electrophysiological platforms. (**A**) Photocurrent density output curves of Te devices at varying laser intensities under NIR-II (1550 nm) stimulation. There are distinct photocurrents under zero bias (as indicated by the red dashed lines) and an increase with higher intensities. (**B**) Photocurrent densities of Te devices varying with laser intensity in different laser bands, including VIS (520 nm), NIR-I (1060 nm), and NIR-II (1550 and 2000 nm). The photocurrents increase linearly with intensity in all bands, and the dashed lines indicate the linear fitting. (**C**) Photocurrent density statistics for different types of materials under zero bias, including 1D [TeNWNs, WS_2 nanotubes (*32*)], 2D [$CuInP_2S_6$ (*33*)], and 3D [$BiFeO_3$ (*34*), $LiNbO_3$ (*35*), KBNNO (*36*), halide (*37*), PLZTN (*38*), TTF-CA (*39*), $BaTiO_3$ (*40*), BBLT (*41*)] materials. As the dimension decreases, photocurrents show an elevated trend. TeNWNs exhibit the highest photocurrent densities available, benefiting from asymmetries due to both external interface effects and internal defects. (**D**) Summary of photosensitivity restored using different materials and mechanisms [Si-PV (*14*–*16*), NW-PV (*42*), capacitive (*43*), WC (*1*), rods-PT (*4*)]. Because of a narrow bandgap, high light absorption, and photoelectric conversion, TeNWNs push the photosensitivity from the VIS to a record NIR-II band (1550 nm, light red region), as illustrated by the red arrow. NW, nanowire; PT, photothermal; PV, photovoltaics; WC, wavelength conversion. (**E**) Schematic of TeNWN photocurrent testing under an electrophysiological platform, which integrates the multielectrode array, digital micromirror device (DMD), and 635 nm laser. A VIS laser (635 nm) was used to illuminate the DMD, which projected patterns onto the surface of TeNWNs. (**F**) Voltage response on the surface of the TeNWNs under a 635-nm laser visual pattern (circle and triangle) illumination. The color distribution is linearly mapped to the amplitude of voltage response on the surface of the TeNWNs. Laser intensity is 42.5 mW mm^{-2}, and the recording area is 1200 μm by 1200 μm. The corresponding projection pattern is shown at the top left.

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Retinal ganglion cell responses induced by TeNWN photocurrent stimulation

To further estimate the threshold for retinal activation, we recorded the photocurrents of TeNWNs under VIS [represented by 635 nm (red)] or NIR-II (represented by 1550 nm) laser stimulation at various intensities and pulse durations in saline solution (fig. S18A). As the laser intensity increased in both the 635-nm and NIR-II 1550-nm ranges, the photocurrent exhibited an intensity-dependent increase, following a steady upward trajectory (fig. S18B). Additionally, as the pulse duration increased, the photocurrent gradually rose and reached a plateau in response to a 1-s VIS stimulation. The photocurrent response to 1550-nm stimulation continued to increase over a pulse duration of up to 4 s (fig. S18C).

To assess the vision restoration capabilities of TeNWNs, we used Pde6b$^{rd1/rd1}$/cDTA mice, a genetic blind-mouse model of inherited retinal degeneration characterized by the loss of all photoreceptors by postnatal day 20 (fig. S19) (*44*). Retinal ganglion cell (RGC) activity was recorded from normal (nonblind) and Pde6b$^{rd1/rd1}$/cDTA blind mice, in which the TeNWNs formed a heterogeneous interface with the retinal cells, as illustrated in Fig. 3A. In blind retinas with TeNWNs, we observed robust intracellular spiking activities from RGCs under VIS (blue, 470 nm; yellow, 570 nm; red, 635 nm), NIR-I (940 nm), and NIR-II (1550 nm) illumination (Fig. 3B and figs. S20, A to D, and S21A, top). By contrast, we did not detect responses in the retina of normal mice at 635, 940, or 1550 nm (fig. S21A, bottom), which is consistent with rodent sensitivity to ultraviolet (358 to 359 nm), blue (400 to 500 nm), and green (509 to 510 nm) light (*45*–*47*).

Next, to investigate the intensity threshold for RGC activation mediated by TeNWN photocurrents, we recorded RGC responses to NIR-II, NIR-I, or VIS (470, 570, and 635 nm) laser stimuli with varying intensities [14.4 to 58.0 mW mm^{-2} for NIR-II (Fig. 3, C and E); 23.4 to 78.4 mW mm^{-2} for NIR-I (Fig. 3E and fig. S20B); 0.8 to 513.6 mW mm^{-2} for 635 nm (Fig. 3, D and E); 113.2 to 415.2 mW mm^{-2} for 570 nm (Fig. 3E and fig. S20C); 295.6 to 784.8 mW mm^{-2} for 470 nm (Fig. 3E and fig. S20D)] and various pulse durations [0.15 to 1.0 s for NIR-II (fig. S21, C and D); 0.5 to 4 s for 635 nm (fig. S21, E and F)] to blind retinas with and without TeNWNs. For each condition, we defined the fraction of responses as the number of successful trials over a total of 10 stimulus trials. The activation threshold (the stimulus conditions producing a 50% fraction of response in RGCs) was 18.98 mW mm^{-2} and 0.18 s for 1550 nm, 83.43 mW mm^{-2} and 0.93 s for 635 nm, 45.74 mW mm^{-2} for 940 nm, 276.8 mW mm^{-2} for 570 nm, and 506.23 mW mm^{-2} for 470 nm (Fig. 3, E and F). The proportion of RGCs responding to different wavelengths was 78.6% for NIR-II (1550 nm), 50% for NIR-I (940 nm), 57.14% for 635 nm, 62.5% for 570 nm, and 64.3% for 470 nm (calculated based on data from Fig. 3E). The latencies to the onset of the stimuli are in the range of 200 to 600 ms, with smaller latencies in response to stimuli with larger light intensities (Fig. 3G). The average latencies at 50% cellular response at the activation threshold (1550 nm: 18.98 mW mm^{-2}; 940 nm: 45.74 mW mm^{-2}; 635 nm: 83.43 mW mm^{-2}; 570 nm: 276.8 mW mm^{-2}; 470 nm: 506.23 mW mm^{-2}) were similar for the different wavelengths tested (fig. S21B). To further examine the temporal features in RGC responses, we used stimuli from 1 to 12 Hz (0.5 duty cycle). Despite the fact that there is a few-hundred-millisecond delay in the onset of response, TeNWN-mediated RGCs seemed to catch up with stimulation frequencies of up to 5 Hz (Fig. 3H and fig. S20E). Altogether, these electrophysiological experiments demonstrate optical-action potential coupling properties of RGCs when stimulated by the photocurrents of TeNWNs.

To evaluate the spatial resolution from TeNWN-attached blind-mice retinas, we mapped out the receptive fields of the electrophysiologically recorded RGCs. Following protocols from previous studies (*15*, *48*, *49*), we recorded TeNWN-mediated RGC responses in a seven-by-seven array and smoothed and thresholded the responses (fig. S22C) to show the location and size of the receptive fields. Figure 3I and fig. S22, D and E, show the thresholded receptive field of 22 RGCs from TeNWN-attached blind-mice retinas and 14 RGCs from normal-mice retinas. The areas of RGC receptive fields were 135.57 square degrees (deg^2) in TeNWN-attached blind-mice retinas, which were similar to those in normal-mice retinas (163.02 deg^2) (Fig. 3J).

We also identified the types of recorded RGCs by biotin-labeled reconstruction of the dendritic field during patch-clamp recordings. Briefly, we defined ON, OFF, and ON-OFF types of RGCs based on the depth distribution of biotin-labeled fluorescence signals from RGC dendrites in the choline acetyltransferase (ChAT)–labeled ON and OFF layers (Fig. 3K and fig. S22, A and B). Seven of 38 RGCs were ON-OFF type, 13 of 38 RGCs were OFF type, and 18 of 38 RGCs were ON type. All labeled RGCs exhibited delayed ON-type responses to light stimuli (Fig. 3L).

Visually evoked potentials in the visual cortex of TeNWN-implanted blind mice

To investigate whether NIR light–induced retinal activity was propagated to higher visual centers, we subretinally implanted the TeNWNs in blind mice (4 to 5 months old) and performed electrophysiological recording to assess visual cortical activity. Before the electrophysiology experiments, we confirmed surgical success and retinal reattachment by fundus images, fundus fluorescein angiography (FFA), and optical coherence tomography (OCT) (Fig. 4, A to C, and fig. S23). B-scan OCT images showed that the TeNWNs were successfully implanted into the subretinal space and likely attached to the inner nuclear layer (INL) of the retina (Fig. 4A and fig. S23, B and C). Figure 4B shows that there was no detectable retinal damage 30 days after the implant surgery. The FFA image shows that the implantation area of the retina had no detectable tissue hemorrhage and microvascular changes 14 days after implantation (Fig. 4C and fig. S23, D and E).

We also conducted flash electroretinogram potential (fEEP) experiments to examine the light-intensity saturation in the dark-adapted condition in TeNWN-mediated vision in blind mice. In normal mice, both a- and b-wave components of fEEP signals exhibited saturated amplitude as the light intensity increased (fig. S24) (*50*, *51*), indicating activation of the available rhodopsins in the photoreceptors. In TeNWN-implanted blind mice, amplitudes of both a- and b-waves increased as the light intensity increased (fig. S24), suggesting that no amplitude saturation of fEEP signals was detected within the safety range of light intensities at 1550 nm [International Electrotechnical Commission (IEC) 60825-1, "Safety of laser products"].

We conducted flash visually evoked potential (fVEP) recordings and plotted the raw traces and also averaged traces of the fVEP signals for each mouse (Fig. 4D and fig. S25, A and B). We observed fVEPs in TeNWN-implanted blind mice under 940-nm (red curve in Fig. 4G) stimulation, though with a difference in the latency to the first negative wave (N1 wave) (Fig. 4, E and F) and a reduced peak-to-peak amplitude relative to normal controls (Fig. 4, E and G). These variations in amplitude and latency may stem from the constrained implantation area and ON-type activation as well as long delays in both morphological ON- and OFF-type RGCs mediated by TeNWNs.

Neurons in the visual cortex tend to fire synchronously according to the frequency of flickering light, known as steady-state visual evoked potential (ssVEP) (*52*, *53*). To record the ssVEP response, full-field flicker stimuli from 1 to 12 Hz were presented to head-fixed mice (fig. S25, A and B). The ssVEP spectrum was quantified using fast Fourier transform, revealing that both normal mice and those implanted with TeNWNs exhibited an optimal response frequency of the ssVEP spectrum that was closely aligned with the stimulation frequency (1, 4, 8, 9, 10, 11, and 12 Hz) (fig. S25, C to E). We observed a similar neural synchrony with flicker frequency in the electroencephalogram (EEG) activity of TeNWN-implanted blind mice, and the temporal resolution of the ssVEP spectrum in TeNWN-implanted blind mice closely matched that of normal mice (fig. S25F). We also found

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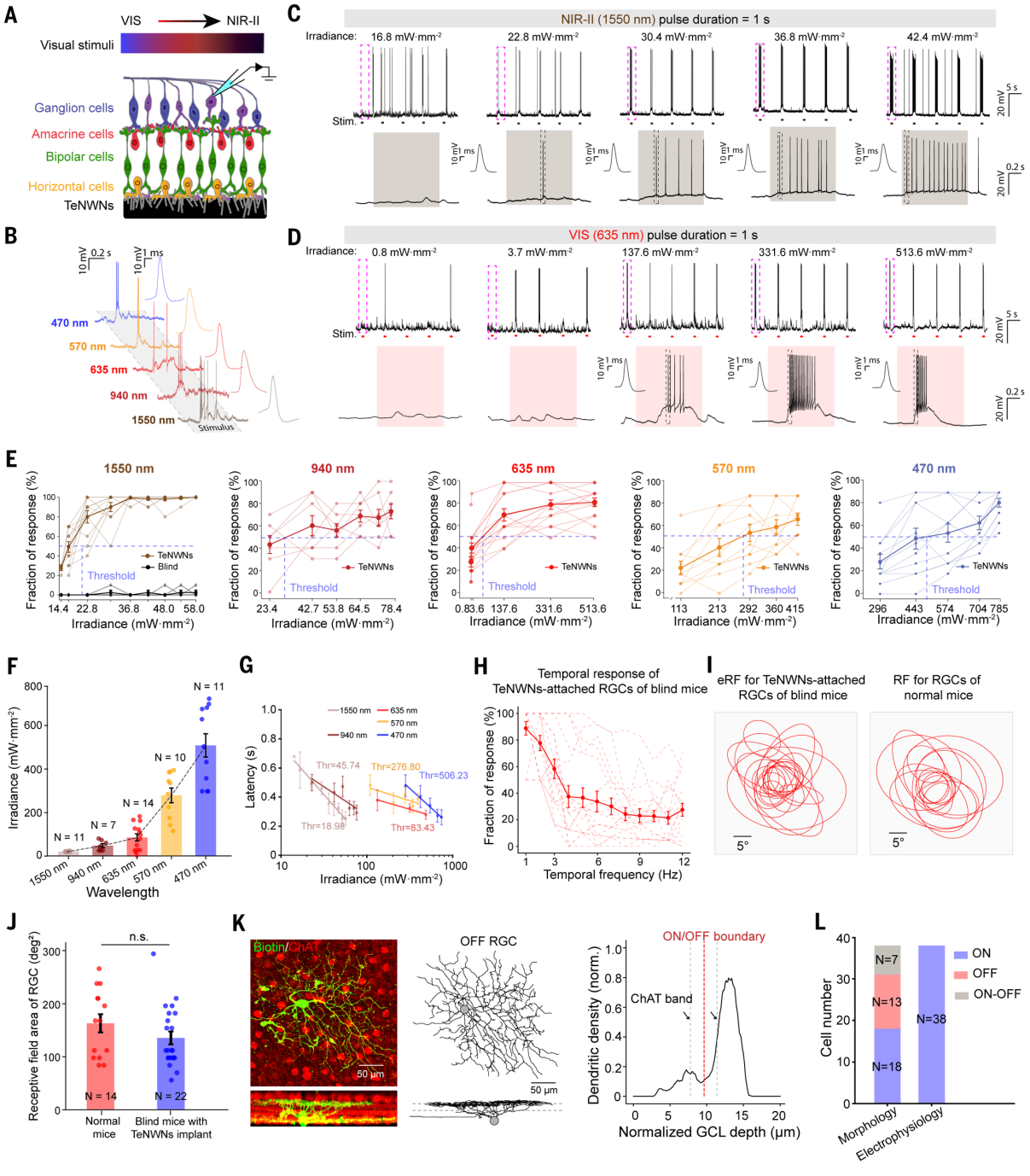


Fig. 3. TeNWN-mediated electrophysiological responses of RGCs from blind-mice retinas. (**A**) Schematic of ex vivo patch-clamp recordings of TeNWN-attached RGCs from blind-mice retinas. (**B**) Representative TeNWN-mediated responses of RGCs to blue (470 nm: 506.23 mW mm^{-2}), yellow (570 nm: 276.8 mW mm^{-2}), red (635 nm: 83.43 mW mm^{-2}), NIR-I (940 nm: 45.74 mW mm^{-2}), and NIR-II (1550nm: 18.98 mW mm^{-2}) illumination. Magnified views of spike waveforms are shown to the right of each image. (**C**) Raw traces of TeNWN-mediated responses of RGCs to 1550-nm stimulation with intensities of 16.8, 22.8, 30.4, 36.8, and 42.4 mW mm^{-2} (top), with corresponding magnified views of raw traces and spike waveforms shown below. (**D**) Raw traces of TeNWN-mediated responses of RGCs to 635-nm stimulation with intensities of 0.8, 3.7, 137.6, 331.6, and 513.6 mW mm^{-2} (top), with corresponding magnified views of raw traces and spike waveforms shown below. (**E**) Average fractions of TeNWN-responsive RGCs under 1550 nm (n =11), 940 nm (n = 7), 635 nm (n = 11), 570 nm (n = 10), and 470 nm (n =11) laser stimulation with different light intensities. The dashed blue lines indicate the stimulation threshold (50% fraction of responses), and light intensities correspond to the stimulation threshold. (**F**) Light intensities with 50% TeNWN-mediated RGC responses. (**G**) Latencies of TeNWN-mediated RGC responses to the onset of the stimuli at different light intensities. (**H**) Fraction of RGC responses in TeNWN-attached blind retinas to flash stimulus of varying temporal frequency from 1 to 12 Hz (n = 18 cells from nine mice). (**I**) Distribution of receptive fields of normal-mice and blind-mice retinas with TeNWNs attached. (**J**) Areas of receptive field in normal-mice retinas (red, n = 14 cells from five mice) and blind-mice retinas with TeNWNs attached (blue, n = 22 cells from 14 mice). (**K**) Shown on the left is a representative whole-mount image of biotin-labeled RGCs (green) and ChAT-labeled cells (red) (top), and the z-stacks of confocal fluorescent images of the RGCs (lower). Shown in the middle are camera lucida drawings of the RGCs (top) and dendritic stratifications (bottom). Gray dashed lines represent ChAT bands (bottom). Shown on the right is the dendritic density of the RGCs (dark). The red dashed line represents ON-OFF boundary. Gray dashed lines mark ChAT bands. (**L**) Cell numbers of ON type (blue), OFF type (red), and ON-OFF type (gray), according to the morphology and physiological response. Data are presented as means ± SEM.

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that the amplitude of the ssVEP response peaked at around 4 Hz and decreased as the frequency increased (fig. S25G). These results indicate that subretinal TeNWN implants restore NIR sensitivity on blind retinas and enable transmission of NIR-driven signals to higher-order visual processing centers.

Biocompatibility

To assess the biocompatibility and biosafety of TeNWNs, we performed whole-mount retinal staining at 30 and 60 days after implantation (fig. S26, A and B). The substrate of the TeNWNs can be sectioned together with the retinal tissue using cryostat, hence the implant area can be easily identified in the retinal slices. We observed a clear elevation at the subretinal implant site (fig. S26, C to E). We quantified the number of both bipolar cells and RGCs at the implant site and compared these values to those of the nonimplanted area at the same eccentricity. We found no difference in the number of RGCs (fig. S27, A to C) or the number of bipolar cells (fig. S28, A to C) between the implanted and nonimplanted areas 7, 14, 30, and 60 days after the implant surgery.

We performed immunostaining against glutamine synthetase (GS; which labels Müller cells) and glial fibrilary acidic protein (GFAP; which labels activated Müller cells) in both the RGC layer and bipolar cell layer. Activated Müller cells labeled by GFAP were slightly more numerous at the implant site than in the area without an implant of the same retina 7 days after the implant surgery (fig. S29, A and B). These slight differences remained 14 days after the surgery (fig. S29, C and D). We did not conduct quantitative analysis on the GFAP-labeled Müller cells because of the large density of GFAP-labeled Müller cells and difficulty in identifying the cell bodies.

We also conducted immunohistochemistry staining of Iba1 (a marker of microglia cells) and calculated the normalized number of Iba1-labled cells in both the RGC layer and bipolar cell layer in whole-mount retina. Similar to GFAP signals, we found a slightly larger number of Iba1-labeled microglia cells at the ganglion cell layer (GCL) of the implanted area compared with the nonimplanted area 7 days after the implant surgery (fig. S28, A, B, and D). We did not find a difference in the number of Iba1-labeled microglia cells 7, 14, 30, or 60 days after the implant in either the GCL or bipolar cell layer (figs. S27, A, B, and D, and S28, A, B, and D). In addition, we conducted TUNEL (terminal deoxynucleotidyl transferase–mediated deoxyuridine triphosphate nick end labeling) staining to reveal the apoptosis in the implanted area, which is comparable to that of control mice 7 days after the implant surgery (fig. S30). These results showed good biocompatibility and low immune response in TeNWN-implanted blind mice.

Behavioral validation of vision restoration and augmentation after TeNWN implantation in blind mice

The pupillary light reflex (PLR) is a well-established method for the primary assessment of light-sensitivity recovery in blind mice (*42*, *54*, *55*) (fig. S31A). To evaluate PLR responses, we set the laser intensity to 836 μW mm^{-2} at 1550 nm, well below the ocular safety threshold for NIR-II light exposure (~4.5 mW mm^{-2} at 1550 nm; IEC 60825-1, "Safety of laser products") (*56*). Both the contralateral and ipsilateral eyes in normal mice showed pupil constriction under white light but no change under VIS red or NIR-II stimulation (fig. S31, B to E). Unlike blind mice, TeNWN-implanted blind mice demonstrated robust pupil constriction both in ipsilateral and contralateral eyes in response to white light, VIS red, and NIR-II light (fig. S31, C to E). These results indicate that TeNWN implantation extended photosensitivity to VIS red and NIR-II light.

Beyond the subconscious PLR responses elicited by TeNWMs, we further investigated the conscious perception of NIR light in TeNWN-implanted blind mice (Fig. 4H). In a behavioral task, mice were trained to lick water within 3 s after light off (correct trial). The correct rate was defined as the ratio of correct trials to total trials ($n = 40$) during the test period. Both normal mice and TeNWN-implanted blind mice exhibited high behavioral accuracy to white light stimulation (normal: 98.33%, TeNWN-implanted: 85.83%; movie S1). However, whereas normal mice failed to detect NIR laser stimulation (correct rate: 11.67%), TeNWN-implanted blind mice demonstrated a higher correct rate (~66.67%) (Fig. 4I). Furthermore, TeNWN-implanted blind mice demonstrated progressive improvement in accuracy from day 1 to day 7 under both NIR and white light stimulation. By contrast, normal mice maintained consistently high accuracy under white light stimulation throughout the experiment (Fig. 4J). This behavioral experiment further validates that TeNWN implantation enabled blind mice to restore VIS light perception while also acquiring enhanced sensitivity to NIR light. These findings also suggest that a training period was still required for TeNWN-implanted blind mice to fully adapt to recovered photosensitivity and achieve performance comparable to that of normal mice.

Next, to determine whether TeNWN-implanted blind mice could localize light sources, we designed a choice-box behavioral test in which mice learned to touch a screen where a light-emitting diode (LED) was lit. A water reward was provided upon successfully touching the target area (a 3-cm radius surrounding the light spot). Both normal mice and TeNWN-implanted blind mice progressively improved in accuracy under white light illumination (Fig. 4K and fig. S31F), suggesting that both groups learned and adapted to the light-source localization task. Whereas the correct rate of TeNWN-implanted blind mice in response to white light stimulation remained slightly lower than that of normal mice, it was higher than that of untreated blind mice (normal: 78.33%, TeNWN-implanted: 67.50%, untreated blind: 26.67%) (Fig. 4K, left, fig. S31F, and movie S2). Moreover, TeNWN-implanted blind mice also demonstrated a higher correct rate in response to NIR (940 nm) light compared with both normal and untreated blind mice (normal: 23.33%, TeNWN-implanted: 71.67%, untreated blind: 25.00%), which were both below chance level (Fig. 4K, right, and fig. S31F). These findings indicate that TeNWN implantation enables blind mice to localize light sources under both white light and NIR illumination.

To further evaluate whether TeNWN-implanted blind mice developed higher-order visual processing capabilities, we conducted a pattern recognition test (Fig. 4L). Mice were trained to selectively respond by touching an illuminated triangular target pattern in randomized trails alternating between triangular and circular light stimuli flashing on the left and right sides of the screen. The accuracy rate was defined as the proportion of trials in which the target pattern was correctly selected out of all trials ($n = 40$) during the test period. TeNWN-implanted blind mice achieved correct rates of ~62.3% for VIS red (635 nm) light stimulation or ~57.75% for NIR (940 nm) light stimulation (Fig. 4, M and N). Meanwhile, correct rates for normal and untreated blind mice were lower than chance for VIS red (movie S3) or NIR light (movie S4). These results suggest that TeNWN-implanted blind mice not only regained light perception but also acquired the capacity for visual discrimination of geometric patterns. Taken together, these results provide compelling evidence that subretinal TeNWN implantation restores broadband photosensitivity and confers functional IR vision in blind mice, with the capacity for both light localization and pattern recognition.

Validation of biocompatibility, safety, and feasibility of TeNWN implantation in a nonhuman primate

We evaluated the biocompatibility, safety, and feasibility of TeNWNs in a nonhuman primate (Fig. 5A and fig. S32). A 2-mm–by–2-mm TeNWN prosthesis on 100-μm-thick mica substrate was implanted into the subretinal space of a *Macaca fascicularis* at eccentricities of 15°. Retinal degeneration was assessed at 7, 28, 67, 90, and 112 days after implantation using fundus photography and OCT, autofluorescence

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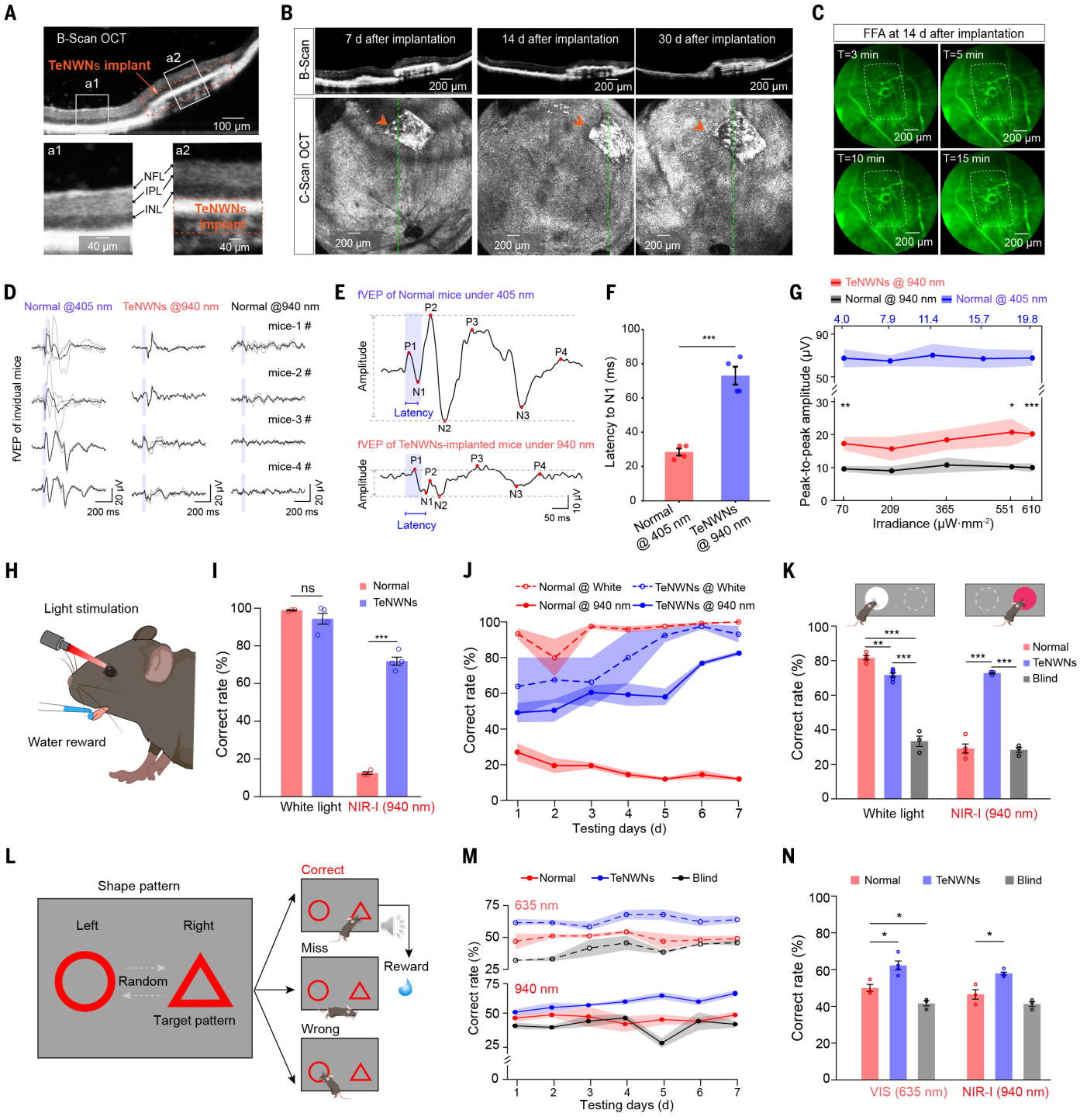


Fig. 4. Restoration and enhancement of visual function in TeNWN-implanted blind mice. (**A**) Representative B-scan OCT images of a retina from a TeNWN-implanted mouse (mouse no. 19). The red dashed box indicates the TeNWN implant. Implantation area (a2) and nonimplantation area (a1) were enlarged in the bottom panel to display the placement of the TeNWNs under the INL of the retina. NFL, nerve fiber layer. (**B**) The B- and C-scans (en face) of the OCT image from mouse no. 19 imaged at 7, 14, and 30 days after implantation surgery. The red arrowheads in the C-scan OCT indicate the position of TeNWNs implant, and the green dashed lines indicate the cross-sectional position of the OCT B-scan. (**C**) FFA of a TeNWN-implanted mouse (mouse no. 19) at different time points (2, 5, 10, and 16 min) after intravenous injection of indocyanine green at 112 days after implantation. The white dashed boxes indicate the position of the TeNWN implant. (**D**) Representative raw traces of fVEPs in normal mice by ultraviolet (left), TeNWN-implanted blind mice by NIR (middle), and normal mice by NIR (right). (**E**) Representative fVEP waveforms from normal mice (top, $n = 4$) and TeNWN-implanted blind mice (bottom, $n = 4$). The red dots indicate the different components of fVEP (P1N1, P1P2, N1P3, and P4). (**F**) The latencies of the N1 wave in normal mice and TeNWN-implanted blind mice. (**G**) The difference between the highest peak and the lowest peak amplitude within 300 ms after the onset of stimulation under different power intensities (right) in TeNWN-implanted blind mice under 940-nm laser stimulation (red lines), normal mice under 940-nm laser stimulation (black lines), and normal mice under 405-nm laser stimulation (blue lines). The shaded regions indicate the 95% confidence intervals. (**H**) Schematic of light-cued water-reward learning. (**I**) Average correct rate of water licking behavior of normal mice (red) and TeNWN-implanted blind mice (blue) under white LED (left) and NIR (940 nm, right) laser stimulation over the testing phase. (**J**) Average daily

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changes in correct rate of water licking behavior of normal mice (red) and TeNWN-implanted blind mice (blue) under white LED (dashed line) and NIR (940 nm, solid line) stimulation during the testing phase from day 1 to day 7. (**K**) Correct rate of light-source location choice-box behavioral test in normal mice (red), blind mice (black), and TeNWN-implanted blind mice (blue) under VIS (635 nm, left) and NIR (940 nm, right) laser stimulation. (**L**) Schematic of the pattern recognition choice-box behavioral test. (**M**) Average daily changes in correct rate of pattern recognition of normal mice (red), blind mice (black), and TeNWN-implanted blind mice (blue) under VIS (635 nm, dashed lines, top) and NIR (940 nm, solid lines, bottom) laser stimulation during the testing phase from day 1 to day 7. The shaded regions indicate the 95% confidence intervals. (**N**) Average correct rate of pattern recognition of normal mice (red), blind mice (black), and TeNWN-implanted blind mice (blue) under VIS (635 nm, left) and NIR (940 nm, right) laser stimulation over the testing phase. ***$P < 0.001$; **$P < 0.01$; *$P < 0.05$; ns, not significant. Data are presented as means ± SEM.

(AF) imaging, and FFA. At 7, 67, and 112 days after implantation, no evident abnormalities were observed on the surface of the implanted eye (fig. S32A). The lens maintained good transparency, with no signs of clouding or exudation (fig. S32A). At 7, 28, 67, 90, and 112 days after implantation, the TeNWNs remained stable and maintained close contact with the INL of the retina (Fig. 5A). To comprehensively assess retinal changes at different time points after prosthesis implantation, we defined a circular region centered on the material surface with a diameter of 1 mm. The annular region between 2.22 and 1 mm was subdivided into four quadrants (a to d). The average thickness of different retinal layers, including the GCL, inner plexiform layer (IPL), INL, outer plexiform layer (OPL), and outer nuclear layer (ONL), was measured at five specific locations within these quadrants based on semiautomated retinal layer segmentation (fig. S32B, top, and fig. S32C). We found a change in thickness at day 28 (fig. S32C), which might be due to unexpected disruption during the acquisition of OCT images (Fig. 5A).

To reveal the vascular distribution at the implant site, we conducted AF imaging 112 days after the implant surgery, which revealed no bleeding at the implant site (fig. S32D). FFA images in the *Macaca fascicularis* showed the intact retinal vasculature overlaying the TeNWN implant 112 days after the surgery (Fig. 5B), providing evidence for vascular changes over time.

To further characterize the functional impact of the implants, we recorded fEEPs in response to NIR light in the implanted eye of a healthy *Macaca fascicularis*. We performed acute recordings using a 500-ms pulse of VIS red (635 nm, 35.71 μW mm^{-2}) or NIR-I light (940 nm, 60.48 μW mm^{-2}), delivered by an LED positioned 2 cm in front of the eye. The eye with the implant maintained responsiveness to VIS red light. Moreover, fEEPs were successfully elicited under NIR-I illumination 19 days after the implant surgery (Fig. 5, C and D). The amplitudes of both the a- and b-waves in the TeNWNs-implanted eye were larger than that of the unimplanted eye in response to the NIR-I illumination 90 days after the implant surgery (Fig. 5, E and F). These results demonstrate that photosensitivity can be augmented to the NIR range by TeNWNs in a nonhuman primate.

Discussion

The human retina has different types of photoreceptors, namely S-, M-, and L-cones, which exhibit spectral sensitivities to blue, green, and red light, respectively. In our study, TeNWNs-mediated broadband vision shows preferred sensitivity to IR light over visible light. Previous studies showed that roughly one-half of murine RGCs are responsive to both short and middle wavelengths, indicating that they receive input from two different types of cones (*57*). Photoreceptors in vertebrate duplex retina have evolved functional specialization, namely, highly sensitive but monochromatic scotopic vision (rods) and highly resolved photopic color vision (cones). The sensitivity of rods is at least 20 times greater than that of cones, whereas the response time for rods is roughly 20 times slower than that of cones in normal-mice retinas (*58*). The response intensity threshold at 470 nm was almost 30 times higher than the threshold at 1550 nm. These response characteristics of the TeNWNs suggested that the artificial retinal prosthesis potentially has "pseudo-scotopic NIR vision" and "pseudo-photopic color vision." Note that the overall sensitivity for the TeNWNs is much smaller than that for the human rods and cones, hence external virtual reality goggles with compensation for sensitivity and spectral selectivity may be necessary to achieve better performance in prosthetic vision.

Previous studies in rd1 mice showed that despite the integrity of the inner retinal layers (*59*), dendrites of RGCs exhibited projections to the distal region of the IPLs (*60*), indicating the plasticity in circuit changes in diseased retina. We also explored the changes in cortical circuit and behavioral performance due to cortical plasticity. Our previous work also showed a gradual improvement in orientation discrimination tasks after blind mice were implanted with titanium oxide nanowires (*49*). In this work, we found that visual cortical neurons in blind mice exhibited enhanced response amplitude and coding capability over the course of 2 months after implantation, indicating a change in the cortical circuity. In general, visual areas seem to retain much less plasticity in adulthood than auditory areas (*61*). Unlike patients with cochlear implants who experience more-natural hearing capability after training, visual prosthetic patients exhibited improvements in visual tasks with simple cues, including locating bright squares and identifying the presence of simple objects (*9*, *62*). Lessons from perceptual learning in sighted individuals suggested that the effectiveness of learning increases with stimulus complexity (*63*), with electrophysiological evidence of top-down influences in primates (*64*). This opens up the possibility that after training with more complex patterns, higher-level visual areas may compensate, though with limited amount, for distortions earlier in the visual pathway. Another observation in blind patients is that motion processing seems to be more robust than shape processing after long-term visual deprivation (*65*), indicating that there may be differential plasticity in distributed visual areas. Specialized design for perceptual learning tasks is necessary to ensure improvements in most desired visual functions for blind patients.

Through rational engineering of material distribution, bandgap, and built-in asymmetries, we have developed a retinal nanoprosthesis using TeNWNs that might be able to restore and extend vision to the IR range. TeNWNs are readily implantable and capable of spontaneously and efficiently converting photons into photocurrents for retinal cell activation. Furthermore, the device achieves exceptionally high photocurrent density up to ~30 A cm^{-2} and an extremely broad photosensitivity range from VIS to an NIR-II wavelength of 1550 nm. Subretinal implantation in blind mice reactivated dormant visual pathways, elicited cortical responses over flicker frequencies of 1 to 12 Hz, and enabled behavioral discrimination tasks under VIS and IR illumination at clinically safe irradiances (100 μW mm^{-2}). Moreover, in a healthy nonhuman primate, the TeNWN implant conferred new IR photosensitivity without disrupting normal vision. Our work suggests that the TeNWN retinal nanoprosthesis might be an effective approach for both restoring lost vision and extending photosensitivity to the IR range. The preclinical validation of TeNWNs in blind mice and primate models is an encouraging step toward restoring VIS and augmented IR vision in blind patients.

Materials and methods

Synthesis and characterization of TeNWNs

TeNWNs were grown by chemical vapor deposition. A ceramic boat containing $SnTe_2$ powder (Sigma-Aldrich, 99.99%) was placed in the middle of a quartz tube, while the mica substrate was placed on another quartz boat with N_2 as the carrier gas. The gas flow rate was 50 sscm. The $SnTe_2$ was heated to 500°C, while the substrate growth

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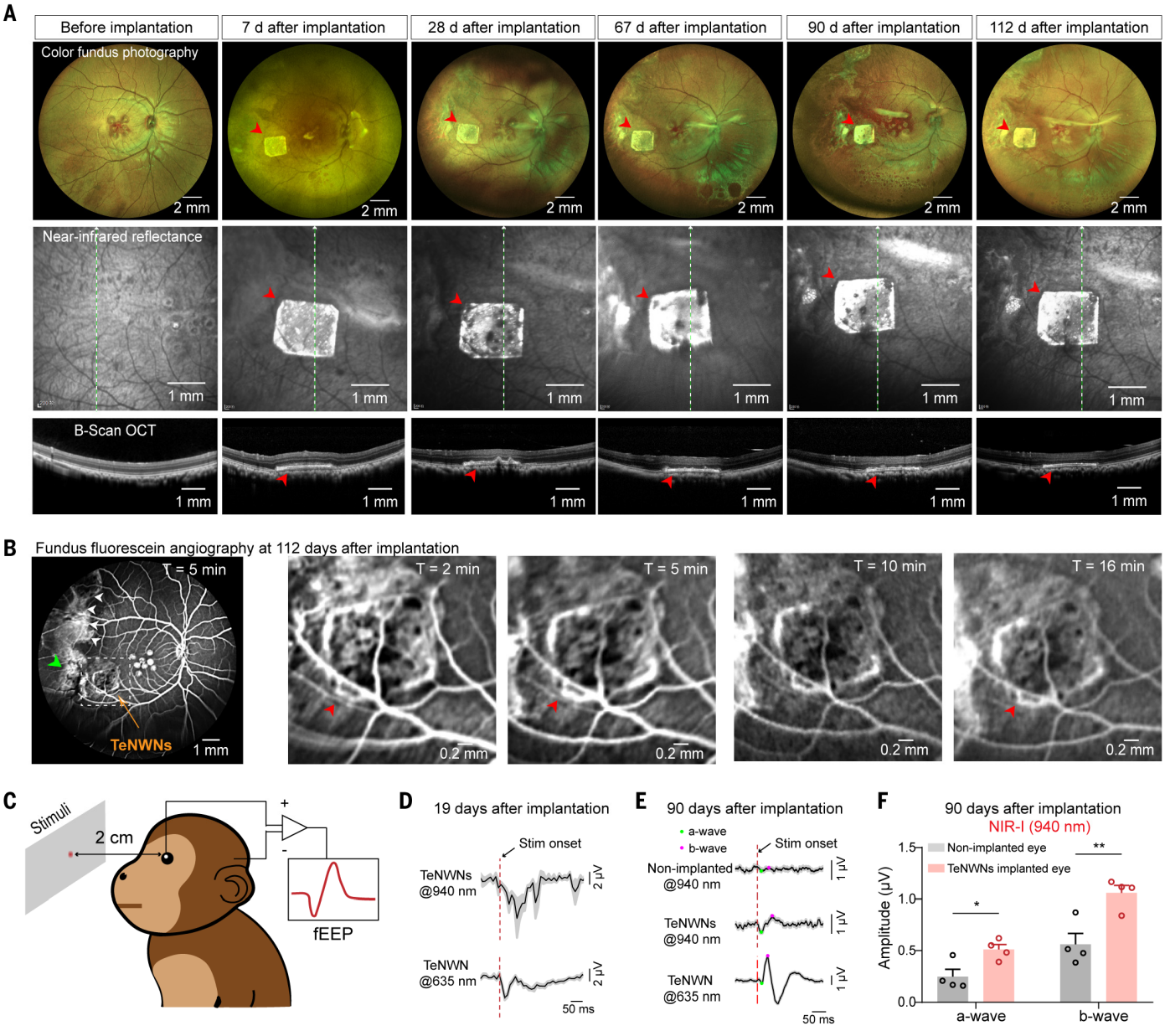


Fig. 5. Ophthalmic assessments and fEEP recordings in TeNWN-implanted *Macaca fascicularis*. (**A**) Fundus photography (100°; top), NIR reflectance imaging (30°; middle), and B-scan OCT (bottom) of the implanted area in the right eye before and at 7, 28, 67, 90, and 112 days after the implant surgery. The green dashed lines indicate the cross-sectional position of the OCT B-scan, and the red arrowheads indicate the position of TeNWNs implant. (**B**) FFA at different time points (2, 5, 10, and 16 min) after intravenous injection of indocyanine green at 112 days after TeNWN implantation. In the image on the left, white arrowheads and white dots in the center indicate surgical sites from a separate study, and the green arrowhead marks the photocoagulation site used to seal retinal openings during the TeNWN implant surgery. In the images on the right, red arrowheads indicate the TeNWN implant. (**C**) Schematic of fEEP recording with NIR-I stimulation.(**D**) Scotopic (dark-adapted) fEEP responses of the TeNWN-implanted eye under NIR-I illumination (top; 500 ms, 940 nm, 60.48 μW mm^{-2}) and VIS light illumination (bottom; 500 ms, 635 nm, 35.71 μW mm^{-2}) at 19 days after TeNWN implantation. (**E**) Scotopic (dark-adapted) fEEP responses of a nonimplanted eye under NIR-I illumination (top; 500 ms, 940 nm, 60.48 μW mm^{-2}) and a TeNWN-implanted eye under NIR-I illumination (middle; 500 ms, 940 nm, 60.48 μW mm^{-2}) and VIS light illumination (bottom; 500 ms, 635 nm, 35.71 μW mm^{-2}) at 90 days after TeNWN implantation. The green and magenta dots indicate the a- and b-wave components of fEEP. (**F**) Amplitude of the a- and b-waves of fEEP at 90 days after TeNWN implantation. Each dot indicates one recording session, and each session contains five trials. Data were analyzed by Student's *t* test. $^{**}P < 0.01$; $^{*}P < 0.05$. Data are presented as means ± SEM.

temperature was 300°C. The growth was held under atmospheric pressure for 30 min, and cooled naturally. It is worth noting that nanowire morphology, defect (vacancy/substitution) properties can be modulated by growth conditions such as temperature (figs. S33 and S34). Morphological characterization of the grown TeNWNs was carried out by optical microscopy (Olympus BX53M) and SEM (ZEISS GeminiSEM300). In order to characterize the lattice structure of TeNWNs, HAADF-STEM was performed along the c-axis cross-section direction. In addition, single nanowire in TeNWNs was characterized by AFM (Oxford Instruments MFP-3D Origin+), showing a thickness of ~150 nm. TeNWNs

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were also characterized by Raman spectroscopy (Semrock Confotec MR200), which showed strong peak at 95, 124, 142 cm^{-1}.

Device simulation and photocurrent calculation

The density functional theory (DFT) calculations are implemented using the Quantum ATK package (*66*). The exchange-correlation functional is treated within the generalized gradient approximation of Becke88-LeeYangParr type (*67*–*69*). The wave functions are expanded as linear combinations of atomic orbitals using norm-conserving pseudopotentials provided by the PseudoDoJo project (*70*). The Brillouin zone is sampled by 17 × 17 × 11 Monkhorst-Pack *k*-point mesh (*71*). The force tolerance on each atom is less than 10^{-2} eV/Å. The Hamiltonian of the two-probe Te device without light is calculated by the DFT coupled with the non-equilibrium Green's function (NEGF) method. In the self-consistent calculation, the k-points are sampled with a separation of about 0.2 $Å^{-1}$ in the Brillouin zone. The photocurrent is calculated as a perturbation to the Hamiltonian caused by the light-matter interaction, following the reported formulas (*72*, *73*). In the simulations, the Te vacancy and Sn substitution defects are introduced by removing the Te atoms or replacing the Te atoms by Sn atoms, respectively. The linear polarized light is shined on the central region. The polarization direction of the light is along the *z*-axis, and the illumination intensity is 0.833 W cm^{-2}.

Device fabrication and experimental photocurrent measurement

Single nanowire in TeNWNs was transferred to a cleaned substrate. The Te optoelectronic device was formed using electron beam lithography (JEOL JSM 7610F) to determine the source and drain electrode patterns, followed by deposition of the electrode metals using electron beam evaporation (DETECH DE400). The probe station (Cascade Summit11000B-M) equipped with semiconductor analyzer (Keysight B1500A) and a laser light source was used to measure the photocurrent of Te device. The laser light source can emit wavelengths ranging from 520 nm to 2000 nm.

Experimental animals and genotyping

The mice were raised and bred in controlled environmental conditions, including a room temperature of 23 ± 2°C, a relative humidity of 60-65%, and a 12-hour light/12-hour dark cycle. The wild-type (C57BL/6J) mice were obtained from the Slac Laboratory Animal Co. in Shanghai, China. The $Pde6b^{rd1/rd1}$/cDTA mice were acquired by crossed the heterozygous $Pde6b^{rd1/+}$ mice mice the with *cDTA* positive mice. The homozygous $Pde6b^{rd1/rd1}$ and heterozygous $Pde6b^{rd1/+}$ gene were identified by forward primer: 5′-CATCCCACCTGAGCTCACAGAAAG-3′ and reverse primer: 5′-GCCTACAACAGAGGAGCTTCTAGC-3′. The homozygous $Pde6b^{rd1/rd1}$ mice displayed two mutant bands at 126 bp and 155 bp, while heterozygous $Pde6b^{rd1/+}$ mice exhibited two mutant bands (126 bp and 155 bp) along with an additional wild-type band at 301 bp. The cDTA knockout mice were identified by forward primer: 5′- CAAGGAAATTATGACGATGATTGG-3′ and reverse primer: 5′-GGCTTGAGCCATATACTCATACATCGC-3′. This primer set produced a mutant band at approximately 450 bp in cDTA knockout mice.

For behavioral experiments, mice aged 4-5 months old were used. For electrophysiological patch-clamp recordings, mice aged 3 months old were used. Most experiments did not specify the gender of the mice used, except for the behavioral experiments, which exclusively used male mice. Mice were randomly assigned to the treatment group and control group. Each mouse was randomly picked up and placed into a separate enclosure before the behavior test. Experimenters were blinded to the animals for behavioral study.

One adult female *Macaca fascicularis*, aged 5.2 years, was housed at InnoStar Bio-tech Co., Ltd, Nantong, China. Following a recovery period of about 7 days from surgery, ophthalmic examinations and scotopic flash electroretinograms (ERGs) were conducted under anesthesia condition. During the procedure, ocular hydration was maintained. After examination, the monkey was returned to a temperature-controlled recovery chamber until fully awake.

Patch-clamp recording of RGCs in mice retinas

Mice were anesthetized with isoflurane at a concentration of 1-2% delivered at a flow rate of 0.5 - 1.0 liters/min. Eyeball of mice was enucleated and immediately placed in oxygenated Ringer's solution (in mM, NaCl 124, KCl 2.5, $CaCl_2$ 2, $MgCl_2$ 2, NaH_2PO_4 1.25, $NaHCO_3$ 26 and glucose 22, pH 7.35; oxygenated with 95% O_2 and 5% CO_2). The retina was carefully separated from the choroid membranes and transferred to one piece of filter paper (Merck Millipore) with a hole in the center. For patch-clamp recording of RGCs, a setup was prepared with the bottom layer consisting of TeNWNs, nanowires facing upwards. Above the TeNWNs, a piece of filter paper with a hole in the center was positioned. The retina was placed on the filter paper with the outer retina facing downwards, making direct contact with TeNWNs through the hole in the filter paper. A metal grid was placed on top of the retina to ensure close contact with the underlying TeNWNs. A glass pipette with a resistance of 5-10 megaohms was pulled using a P-97 micropipette puller (Sutter Instruments) and filled with an internal solution containing various components (in mM), including potassium gluconate 105, KCl 5, $CaCl_2$ 0.5, $MgCl_2$ 2, EGTA 5, HEPES 10, Mg-ATP 4, GTP-Na 0.5, and sodium phosphocreatine 7, adjusted to a pH of 7.4.

Visual stimuli were provided by a setup consisting of two upright DIC microscopes (BX51WI, Olympus; Examiner A1, Zeiss), two DMD systems (FC-LASDLP, Fluoca Scientific) connected to microscopes, multiple laser light sources, and an LED light source (FC904pro, Fluoca Scientific) for different wavelengths. VIS violet (405 nm laser, MDL-MD-405-4W, CNI Optoelectronics), VIS blue (470 nm LED, FWHM 24 nm, FC-904Pro, Fluoca Scientific) VIS yellow (570 nm LED, FWHM 98 nm, FC-904Pro, Fluoca Scientific), VIS red (635 nm laser, MDL-D-635-2W, CNI Optoelectronics), NIR-I (940 nm laser, LP940-SF30, Thorlabs) and NIR-II (1532 nm laser, MDL-N-1532-3.5W, CNI Optoelectronics) light were generated and brought to cells through DIC microscopes by ×10 dry objectives or ×40 water dipping objectives, but in specific experiments through DMD first and then into microscopes. To investigate the response of RGCs to different light intensities, stimulation frequencies and pulse durations, VIS red laser with DMD system, NIR-II laser (1550 nm), NIR-I laser (940 nm), 570 nm LED, 470 nm LED were employed in the relevant experiments. For red light, a light intensity of 0.18, 0.78, 3.56, 36.54, and 137.6 mW mm^{-2} was respectively presented with a set duration of 1 s, while a pulse duration of a 0.5, 1, 2, 3, and 4 s was respectively presented with a set intensity of 137.6 mW mm^{-2}. The illumination area was a 0.363 mm×0.204 mm rectangle with stimulated cell set to its geological center. In the case of NIR-II, a light intensity of 14.4, 16.8, 22.8, 30.4, 36.88, 42.4, 48.0, 53.2, and 58 mW mm^{-2} was respectively presented with a set duration of 1 s; and a pulse duration of a 0.15, 0.20, 0.30, 0.50, and 1 s was respectively presented with a set intensity of 48 mW mm^{-2}. In all cases, 10 stimulations (trials) were provided and repeated for three times for each RGC. The illumination area was a circle with a diameter of 0.575 mm, with stimulated cell set to its center as well. A responsive trial was defined when action potentials were elicited during the stimulation. The fraction of response was calculated as the mean number of responsive trials divided by 10 trials.

To depict the RGC receptive fields of normal retinas and of blind retinas induced by TeNWNs, VIS violet laser with DMD system and VIS red laser with DMD system were employed in the relevant experiments. For both sets of experiments, the visual stimuli used were generated by Psychopy toolbox. A circular light spot with a diameter of 0.136 mm was flashed on a black background in a 7×7 grid in a pseudorandom order. The grid was 0.506 mm×0.506 mm with centers of the circles located on the mesh nodes of the grid, thus there would be overlapping areas between adjacent circles. In all cases,

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the 49 stimulations were presented for 0.5 s followed by a 5 s interval and repeated 10 times for each RGC. For violet light, the light intensity used in the experiments was 37.07 mW mm^{-2}; for red light, the light intensity used in the experiment was 523.03 mW mm^{-2}.

To characterize the temporal response properties of TeNWNs-attached RGCs from blind mice, we employed a full-screen 635 nm laser stimulus under a ×40 water-immersion objective. Temporal flicker stimuli were delivered at frequencies ranging from 1 to 12 Hz, with each frequency presented for a duration of 2 s and a duty cycle of 50%. The light intensity of the 635 nm laser used in this experiment was maintained at 513.6 mW mm^{-2}.

Patch electrophysiology data were collected using a MultiClamp 700B patch-clamp amplifier (Molecular Devices, San Jose, USA) and a Digidata 1440 digitizer (Molecular Devices, San Jose, USA), all under the observation of a DIC microscope (Zeiss, Oberkochen, Germany). The acquired data were analyzed using Clampfit (Axon) to study the action potentials of RGCs. CED spike2 v7.16 (Cambridge Electronic Design, UK) and customized python code were also used. Prism v9.5 (GraphPad) was used to analyze the fraction of responses. CED spike2 v7.16 (Cambridge Electronic Design, UK) and customized python code.

Patch-clamp data analysis

To calculate the receptive field (RF) area for individual RGCs, we defined a pre-stimulus period as the 0.5-s window preceding the stimulus onset to establish the baseline spike rate. The relative spike rate of each RGC recorded by patch-clamp ex vivo was calculated by subtracting the baseline spike rate from the spike rate during the stimulus period. The average spike rate was obtained by averaging across 10 repeats at each of the 49 stimulus locations and then smoothed using a Gaussian filter with a standard deviation (σ) of 1.5 (*48*). The smoothed receptive field was thresholded at the value of the maximum response minus the standard deviation of the response [max(RF) − std(RF)]. The RF center was identified as the center of mass of the largest contiguous region exceeding the threshold, and the RF area was quantified by multiplying the pixel-wise area of this region by a conversion factor of 11.8 deg^2 for each grid.

For temporal feature analysis, a trial was considered responsive if action potentials were elicited during the pulse stimulation at each frequency. The response fraction was calculated as the ratio of the number of responsive trials to the total number of trials. Cells were excluded from the analysis if they did not maintain viability throughout the duration of the experiment.

TeNWN implant surgery and fundus examination in mice

The $Pde6b^{rd1/rd1}$/cDTA mice, which were blind mice aged 8 to 10 weeks, were subjected to the following surgery procedures. First, the blind mice were anesthetized using isoflurane at a concentration of 1-2% delivered at a rate of 0.5 - 1.0 liters/min. The eyeball was positioned appropriately for implantation by suturing the upper and lower eyelids. A small incision was made on the dorsal sclera of the eye. Subsequently, retinal detachment was induced by performing a subretinal injection of 1 to 2 μl of a 0.9% sodium chloride solution using a NanoJect II device (Drummond Scientific Company, USA). The initial incision was expanded to create sufficient space for the implantation of TeNWNs. The area covered by the TeNWNs ranged from 0.5 to 1 mm^2. The TeNWNs, which had been sterilized prior to implantation, were held by tweezer and inserted into the subretinal space at least 1 mm away from the incision site. After implantation, the sutures used for eyeball fixation were carefully removed. The eyeball was washed using 0.9% saline solution. Finally, erythromycin eye ointment was applied to the eye following the procedure. One weeks after the implantation, the placement and eccentricity of TeNWNs in the subretinal space was verified by OCT at 4, 7, 14, and 30 days after surgery (Ultramicro Ophthalmol Imaging System, Optoprobe Science) and fluorescein angiography images with detailed cross-sectional presentations of the implant and the retina. For fundus fluorescein angiography of TeNWNs-implanted mice, we administered 2 ml/kg (10%) indocyanine green via tail vein injection. Fundus imaging was performed using a retinal imaging system (Ultramicro Ophthalmol Imaging System, Optoprobe Science) at 3, 5, 10, and 15 min post-injection, with retinal fluorescence signals excited using a blue laser, while color fundus photographs were captured under white light illumination.

fVEPs and ssVEP recording and analysis in mice

To record the visual evoked potentials of TeNWN-implanted mice in response to light stimuli, we used a custom-made electrode consisting of three stainless steel screws, following our previous experimental setup. The reference electrode was positioned above the frontal cortex (AP = 1 mm, ML = +1.5 mm), while the active electrode was placed above both the ipsilateral and contralateral visual cortex as well as the parietal lobes (AP = −3.5 mm, ML = 2.5 mm) on the skull surface. Prior to fVEP recording, mice were head-fixed on a floating ball system, allowing free movement of the body while maintaining stable head. fVEP recordings were collected from all animals 2-weeks post-implantation. We measured modulation of the cortical response using a gradient irradiance range spanning from 70~610 μW mm^{-2} for 940 nm and 4.0-19.8 μW mm^{-2} for 405 nm. Each averaged fVEP consisted of a total of 200 consecutive light flash, each lasting 50 ms and delivered at a frequency of one flash per second. For ssVEP recording, the full visual field was lit by the light of the one LED. An Arduino Uno microcontroller orchestrated the modulation of LED at all frequencies from 1 to 14 Hz. Each recording session contained 50 trials, wherein each trial featured a precisely timed 2-s interval of flicker stimulation, followed by a 2-s interstimulus interval. Both fVEP signals and ssVEP were recorded by Meidusa system (Medusa, Bio-Signal Technologies, China) at a 500 Hz sampling rate using a 0.5 Hz to 200 Hz bandpass filter. We defined the VEP amplitude as the peak-to-peak variation of the signal during the first 300 ms after the stimulation. The preference frequency domain of EEG response from different group were calculated from a fast Fourier transform (FFT) performed on the averaged ssVEP.

Multielectrode array recording of photocurrents

To assess the spatial resolution of the electrical field in TeNWNs, a section of the material (2×2 mm^2) was placed flat onto the MaxOne high-density (HD) multielectrode arrays (Maxwell Biosystems no. PSM, MEA) a saline solution. A VIS laser (635 nm) with an irradiance of 42.5 mW mm^{-2} was used to illuminate a digital micromirror device (DMD), which projected patterns onto the TeNWNs, as depicted in the schematic (Fig. 2F). The projection area measured 1.2 × 1.2 mm^2. The projection patterns included handwritten digits "0" to "9" from the MNIST dataset (http://yann.lecun.com/exdb/mnist/) (*74*), the capital letter "E" representing different visual angles on the retina, as well as triangle and circle shapes. Voltage waveforms generated by light stimulation were recorded from each of the 1015 electrodes on the MEA at a 20 kHz sampling rate, then amplified and digitized using a custom-built readout and data acquisition system. The voltage amplitudes of individual units were fitted using a Gaussian distribution and mapped according to their corresponding electrode positions on the chip (Fig. 2F and fig. S16).

TUNEL staining

For TUNEL staining, mice were euthanized under deep anesthesia induced by a 2.5% Avertin/saline solution, followed by transcardial perfusion with 0.1M PBS and fixation with 4% ice-cold paraformaldehyde (w/v) in PBS. The dissected retinal tissues were post-fixed overnight in 4% paraformaldehyde at 4°C, cryoprotected in 30% sucrose in 1× PBS at 4°C for 48 hours, and sectioned into 20 μm slices using a Leica CM520 freezing microtome. TUNEL staining was performed using the

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Click-iT Plus TUNEL assay kit (Thermo Fisher Scientific, MA, USA, C10617). Sections were fixed, permeabilized, and incubated with TdT reaction mix containing EdUTP for 60 min at 37°C. After washing, the sections were incubated with a click chemistry reaction mix [Alexa Fluor 488 picolyl azide and Cu(I)] for 30 min at room temperature. After PBS washing, sections were counterstained with DAPI, mounted, and imaged using a FV-3000 (Olympus microscope). TUNEL-positive cells were quantified and analyzed using ImageJ.

Whole-mount retina immunostaining

Implanted mice were deeply anesthetized using an overdose of isoflurane for euthanasia. They were then subjected to transcardial perfusion with 0.9% saline, followed by perfusion with 4% paraformaldehyde (PFA). Implanted eyes were then removed and enucleated, with retina-TeNWN complex carefully dissected from the eye cups. For morphological analysis of patch-clamp recorded cells, these cells were recorded by patch pipettes containing internal solution with the addition of neurobiotin (Vector Laboratories, SP-1120, 4% w/v). The whole-cell patch clamp status was maintained for more than 20 min to ensure diffusion of the tracer throughout the cells. After the recording was completed, the retinas were removed from Ringer's solution with filter paper attached to maintain their surface flatness during the fixation process. All retinas mentioned above were then fixed in 4% PFA for a duration of 5 to 7 hours at a temperature of 4°C and were washed with a Tris-buffered saline (TBS) solution. The retinas were then blocked with a solution containing 10% Donkey serum. After blocking, primary antibodies (anti-Brn3a from Santa Cruz Biotechnology, California, USA, SC-31984, 1:200; mouse anti-PKC-α from Thermo Fisher, Waltham, USA, MA1-157, 1:200; rabbit anti-Iba1 from WAKO/019-19741, JAPAN, 019-19741, 1:800; goat anti-ChAT, Sigma-Aldrich, AB144P, 1:250; mouse anti-GS, #MAB302, 1:2000 dilution, Milipore; anti-GFAF, G6171, 1:400, Sigma Aldrich; mouse anti-rhodopsin, MCA-B630, 1:300, Encor Biotechnology Inc; PNA dye,RL-1072-5, 1:1000,Vector Laboratories) were applied to the retinas according to specific research needs and left to incubate for 24 to 30 hours at 4°C. Then the retinas were exposed to a secondary antibody solution (Donkey anti-goat conjugated to Alexa Fluor 594 and anti-mouse conjugated to Alexa Fluor 488 from Jackson ImmunoResearch, USA, 1:300) or secondary antibody solution added with streptavidin conjugated to alexa-fluor 488 (Invitrogen, S32354, 1:200) for 3 hours at room temperature in the dark. The whole-mounted retinas were then stained with 1:3000 DAPI and rinsed. Finally, these retinas were air-dried and mounted using Fluoromount-G from Southern Biotech, USA. Fluorescent images were captured using a fluorescence imaging microscope (Eclipse Ni, Nikon, Tokyo, Japan) and a confocal multi-photon scanning microscope (AIR-MP, Nikon, Tokyo, Japan). The images were subsequently analyzed using ImageJ software version 1.48v (NIH) and NIS-Elements AR software version 4.30.01 (Nikon, Tokyo, Japan).

Cell counting

Apoptotic cells were quantified by manually counting TUNEL-positive signals in the INL. To ensure representative sampling, three randomly selected rectangular regions within the ONL of the implanted area were analyzed from retinal sections. The number of DAPI-stained nuclei in these regions was used to calculate the average INL cell size. The ratio of TUNEL-positive cells to DAPI-stained cells in the INL was determined for both experimental and control groups, with normalization performed using the non-implanted region as a reference.

To assess inflammation, whole-mount retinas from TeNWN-implanted blind mice were immunostained with the Iba1 antibody to identify microglial cells. Microglial cell counts were performed in three randomly selected regions directly above the TeNWN implant. Additionally, three corresponding regions at the same eccentricity in the non-implanted area were also randomly selected for comparison. Microglial cells were quantified separately in the INL and GCL. The INL was identified based on PKCα-positive cell layer, while the GCL was identified using Brn3a-positive cell layer.

Morphology reconstruction and analysis

In order to acquire dendritic skeleton images of the recorded cells, Z-stack images of cells were filmed by procedures described in the former section and processed by FIJI. Images were first interpolated so that the z-axis would have the same micron per pixel value as the x-axis and the y-axis. Next, the z-stacks were resliced from left to right to turn the yz-plane of the stacks en face. ChAT bands used as references of the analysis were generated by performing standard deviation z-projection of such stacks. The dendrite skeleton images were generated by tracing dendrites semi-automatically on the neurobiotin (488 nm) channel of the interpolated stack in FIJI using the SNT plugin by Arshadi *et al.* (*75*).

Quantified morphological analysis was conducted on the z-projection images of ChAT bands and yz-plane snapshots of the skeleton images mentioned above. ChAT band locations were determined manually by drawing a horizontal line that crossed the center of reconstructed cell body and intersected with two ChAT bands. Due to the well-maintained flatness of the whole-mount retinas, the exact coordinates of the bands were defined as the coordinates of the brightest spot on the horizontal line within each band. According to the arbor segregation principles of the Eyewire Museum by Bae *et al.* (*76*), we assumed that the location of the OFF and ON ChAT band signal maxima could be normalized as 0.28 and 0.65. The ON-OFF boundary of IPL was therefore defined as the physical depth that could be normalized as 0.47, whereas the border of INL and GCL should be at the physical depth normalized as 0 and 1. Finally, the yz-plane snapshot of the skeleton image was binarized and averaged to the z-axis to generate a depth-dendrite density curve. ON-OFF cells that have two obvious peaks separated by the ON-OFF boundary were picked out before analysis, and the ON/OFF property of the rest was defined by the coefficient $C_{ON/OFF}=\frac{AUC_a}{AUC_a+AUC_b}$, where AUC_a stands for the area under curve between the border of GCL and the ON-OFF boundary (sublamina a), and AUC_b stands for the area under curve between the border of INL and the ON-OFF boundary (sublamina b). An ON-RGC was defined as a cell with $C_{(ON/OFF)}>0.5$, while an OFF-RGC was defined as a cell with $C_{(ON/OFF)}< 0.5$.

Pupillary constriction reflex response

Head bars were surgically implanted onto the skulls of mice before conducting behavioral experiments. After a 2-hour period of dark adaptation, fully conscious mice were immobilized by attaching them to the head bars, all while being illuminated with infrared light. For TeNWNs implanted blind mice, light stimuli were delivered to the eye with implantation and the light reflex of the contralateral eye was recorded simultaneously by a near-infrared camera (JAI, Denmark). For blind mice without implantation and normal mice, light stimuli were delivered to one eye and the reflex of the contralateral eye was recorded with the same procedure. Light stimuli included a white LED, as well as 650 nm red light and 1550 nm NIR-II light emitted from a monochrome laser, respectively. The light intensity of the white LED was 100 μW mm^{-2}, while that of the 635 nm laser was 360.14 μW mm^{-2} and the 1550 nm laser was 836 μW mm^{-2}. Each recording session lasted for 40 s, beginning 5 s before a 20 s light stimulus was introduced. Based on the pupil area before the onset of stimulation, the minimum pupil area after the onset of stimulation was normalized. Then the ratio of area reduction of the pupils in three types of mice was calculated under white, red, and infrared light stimulation. The area of the pupils was measured using ImageJ from NIH in the USA. The percentage of pupil constriction was calculated using the formula of $\left(\frac{A_min}{A_pre}\right)\times 100\%$, where *A_min* is the minimum pupil area during light exposure, and *A_pre* is the pupil area immediately before light

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exposure. No mice were excluded from participating in the behavioral task during this study.

fEEP recording and analysis for mice

The fEEP was performed following the protocol described by Kinoshita *et al.* (77). Prior to recording, TeNWNs-implanted and control mice underwent overnight dark adaptation. Under general anesthesia, compound tropicamide eye drops (Santen Pharmaceutical Co., Ltd., Shiga Plant, Japan) were applied to induce pupil dilation. A circular silver wire corneal electrode was positioned on the corneal surface, while a silver needle reference electrode was placed subcutaneously near the eye socket. The ground electrode, also a silver needle, was inserted subcutaneously into the tail. ERG signals were amplified using a Brownlee Precision Model 410 amplifier (USA) at a sampling rate of 1000 Hz. Light stimuli were delivered using a 405 nm laser beam for control mice and a 1550 nm laser beam for TeNWNs-implanted mice and controlled by self-written Arduino code. Each session contains 20 stimuli, which lasts for 500 ms and is separated by 10 intervals. The gradient irradiance for 405 nm was 0.035μW mm^{-2}, 0.14 μW mm^{-2}, 0.243 μW mm^{-2}, 0.547 μW mm^{-2}, 1.081 μW mm^{-2}, 2.902 μW mm^{-2} and 0.003 mW mm^{-2}, 0.006 mW mm^{-2}, 2.9 mW mm^{-2}, 7.3 mW mm^{-2}, 11.3 mW mm^{-2}, 19.7, 27.9 mW mm^{-2} for 1550 nm. Stimulation was controlled via a custom Arduino-based program. Each recording session consisted of 20 stimulus presentations, each lasting 500 ms, with an inter-stimulus interval of 10 s. As a control, the electrode made direct contact with the TeNWNs array under different irradiance conditions. To maintain corneal hydration throughout the recording, 0.3% sodium hyaluronate eye drops (Santen Pharmaceutical Co., Ltd., Shiga Plant, Japan) were periodically applied.

Visually cued associative learning

Mice with head bars underwent a period of water restriction for 3 days prior to their training. During this adaptation stage, they were secured on an air float for 10 min each day. Subsequently, they underwent a 3-day training period followed by a 7-day testing period. Throughout both training and testing, the mice were exposed to light stimuli emitted by either a white LED or a 940 nm NIR light LED for a duration of 1 s each trial. The light intensity of all LEDs ranged from 80 to 100 μW mm^{-2}. The mice were trained to lick water from the water outlet right in front of their mouths upon the presentation of light stimuli. For each trial, water was dispensed 100 ms after the initiation of the light stimulation. During the training stage, 40 trials were conducted daily. This was followed by the testing stage, wherein a trial was considered correct, and a water reward was provided, only if the mice licked the water outlet subsequent to the light stimulation. In cases where the mice did not exhibit the expected response of licking water upon exposure to light, such trials were categorized as errors, and no reward was dispensed. During the testing stage, 40 trials were carried out each day, and the correct rate (defined as the ratio of correct trials to the total number of trials) for water licking was calculated for each day.

Choice-box behavioral test

A custom-made choice-box measuring 240 mm×240 mm×180 mm was constructed using acrylic sheets. The choice-box was equipped with a water delivery system consisting of a water pipe, water port, and water pump provided by Kamoer in Shanghai, China. Additionally, an audio signal system featuring a buzzer was positioned at the center of the choice-box lid. Visual stimuli were placed at two separates position, spaced 12 cm apart. The light intensity of all visual stimuli used in this behavioral task was quantified at a distance of 6 cm from the light source. The visual stimuli were from white LEDs and 940 nm LEDs for separate experiments. The light intensity of all LEDs ranged from 8 to 10 mW cm^{-2}. For pattern discrimination, rounded and triangular line light source was placed right or left randomly across trials in each session. Each line source was made of 24 LEDs (5 mW) of NIR (940 nm) or VIS red (635 nm) irradiation. The intensity of NIR light pattern was about 30-50 μW mm^{-2} and VIS red light pattern was 32.2 μW mm^{-2} measured at the water outlet. Visual stimuli, water delivery system and the audio signal system were all controlled by an Arduino system. All training and testing sessions were acquired by infrared camera and analyzed manually.

The choice-box-based behavioral test consisted of 3 consecutive stages: pre-training trials, training trials and testing trials. The mice underwent a period of water restriction for 3 days before the test was initiated. During the pre-training trials, the mice were allowed to explore the choice box freely 3 days, with a duration of more than 20 min each day to get accustomed to the setup and spatial arrangement of the choice box. If the mice did not interact with the screen, the visual stimulation would persist for 20 s in training trials and 10 s in testing trials. If the mice touched the screen during the visual stimulation, the visual stimulation would be terminated. Mice were first trained to associate audio cues with water reward. Water reward would be given at the water port with simultaneous audio cues regardless of the reactions of the mice, so that the mice might couple water reward and the audio cues together. The next step was to associate the target area touching with water reward. Mice would frequently touch the walls of the choice box during free exploration. Once the visual stimulation locations were randomly touched, an audio cue and water reward would be given to instruct the mice on screen-touching. The steps above were designed in such a manner that the blind mice were also able to complete them. The mice were then sent to the training stage to establish correct connections between the appropriate visual stimulus and audio cues and water reward. Blind mice could not learn the task at this stage, but some of them would continue to touch the screen randomly due to the training of the first stage, which resulted in the correct rate of the blind mice shown in figures. Besides, another proportion of mice would completely abandon pressing the screen and remain still at the water port after multiple failure during the training stage. This led to a rebound in miss rates of those mice and final correct rates below 50% during the testing stage.

Each trial was initiated with an audio signal cue (tone 1). After a 3 s delay, the visual stimulus randomly appeared at one of the two designated visual stimulation locations. The mice were trained to touch the target area, defined by a circular region centered at the LED with a diameter of 6 cm in the light localization test, using their paws or noses. If the mice correctly selected the appropriate visual stimulus within the stimulation duration, an additional audio signal (tone 2) would be emitted by the buzzer, and the mice would be rewarded with access to a drop of water. Consequently, this trial was classified as a correct trial. In contrast, if the mice failed to interact with the screen within the stimulation duration, the trial was considered a miss trial. If the mice touched an area outside the correct visual stimuli region on the screen during the stimulation duration, the trial was considered as a wrong trial. In the cases of both wrong and miss trials, neither the audio cue (tone 2) nor the water reward would be given. The next trial would commence once the mice had moved away from the visual stimulation region.

Once the miss rate among normal mice dropped to less than 50% consistently over a period of 3 consecutive days during the training trials, the behavioral paradigm for all mice was transitioned to the testing trials. No mice were excluded from participating in the behavioral task during this study. The protocol employed in the testing trials closely resembled that of the training trials with some key differences. Specifically, the stimulation duration was shortened from 20 s to 10 s, and the interval between each trial was set to 5 s.

Surgical procedures for TeNWN implantation in *Macaca fascicularis*

The implantation of TeNWNs into *Macaca fascicularis* was conducted at Enos Biotechnology Nantong Co., Ltd., Shanghai. Anesthesia and surgical procedures followed previously established protocols for non-human

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primates (*49*). A small bleb was created in the sub-retinal space near the macular region by injecting an appropriate amount of intraocular infusion fluid with a 38-gauge needle. A retinal incision was then made using retinal scissors. To ensure smooth delivery of the 2×2 mm^2 TeNWN array into the eye, a 2 mm-wide incision was made at the corneal limbus using a 15° side-port blade. The implant was then carefully held with internal limiting membrane forceps and inserted through the corneal limbal incision, guided into the sub-retinal space through the retinal opening. To reattach the detached retina at the implant site, perfluorocarbon liquid (Bausch & Lomb) and silicone oil (Arciolane 3000, Arcadophta SARL) were introduced. Retinal laser photocoagulation was performed along the retinotomy borders to stabilize the implant and prevent further retinal detachment. Finally, the corneal limbal incision was sutured using 10-0 ophthalmic sutures, and postoperative care included the administration of tobramycin-dexamethasone eye drops twice daily for one week.

Fundus image, OCT, FFA, and fEEP recording in *Macaca fascicularis*

The animals were initially sedated via intramuscular injection of ketamine (10 mg/kg) and subsequently anesthetized with intravenous administration of pentobarbital (30 mg/kg) to ensure full anesthesia. Pupillary dilation was induced using Compound Tropicamide Eye Drops (Santen Pharmaceutical) one hour before the examination. Once anesthetized, the animals were positioned prone in a supporting apparatus for imaging. Fundus photography (100°), red-free model fundus imaging (100°) and fundus autofluorescence (30°) were captured by Confocal Laser fundus Imaging Machine (APOLLO) - Model CRO (Suzhou Weiqing Medical Equipment Co., LTD, China), and optical coherence tomography (OCT) was conducted with a Heidelberg Spectralis OCT system (Heidelberg Engineering, Germany). Color fundus imaging, red-free fundus photography, and OCT scans were carried out at baseline (pre-implantation) and at 7, 28, 67, 90 and 112 days post-implantation to monitor retinal conditions over time. For FFA imaging, the general anesthesia on *Macaca fascicularis* used 1% tropicamide and 2.5% phenylephrine eye drops (Santen Pharmaceutical Co., Ltd., Japan) for 15 min before imaging. The ocular surface was kept hydrated using 0.3% sodium hyaluronate eye drops to prevent corneal desiccation. A 20% fluorescein sodium solution (Alcon, USA) was administered intravenously at a dose of 0.1 mL/kg through the saphenous vein. Fundus imaging was performed using a Heidelberg Spectralis HRA (Heidelberg Engineering, Germany). Image acquisition started immediately after injection and continued for 10-15 min to capture different phases of angiography, including early arterial, arteriovenous, venous, and late-phase images. After the procedure, the monkey was monitored until full recovery from anesthesia. Artificial tears were applied as needed, and any signs of adverse reactions, such as hypersensitivity to fluorescein or prolonged sedation, were recorded and managed accordingly.

Following general anesthesia, scotopic fEEPs were recorded in a dark environment after 30 min of dark adaptation. A circular silver corneal electrode was placed on the surface of the cornea, while a silver needle reference electrode was inserted subcutaneously near the eye socket. A ground electrode was placed subcutaneously at the base of the tail. ERG signals were digitized at 1000 Hz, amplified using a Brownlee Precision Model 1440 amplifier (United States), and bandpass filtered between 1 and 100 Hz. Light stimuli, including red (635 nm) and NIR (940 nm) LED flashes, were controlled via custom Arduino code. Each session consisted of 20 light stimuli, with each stimulus lasting 500 ms and separated by an interval of 5 s. To prevent corneal dryness, 0.3% sodium hyaluronate eye drops (Santen Pharmaceutical Co., Ltd, Shiga Plant, Japan) were applied to the cornea throughout the recording sessions.

Retinal thickness analysis

To quantify the retinal changes at different time points following prosthesis implantation, measured the retinal thickness above the TeNWN implant by defined a circular region centered on the material surface with a diameter of 1 mm (e). The annular region between 2.22 mm and 1 mm was subdivided into four quadrants (a, b, c, d). The average thickness among the five point of different retinal layers, including the GCL, IPL, INL, OPL, and ONL, was measured at five specific locations within these quadrants based on semiautomated retinal layer segmentation.

Statistical analysis

No statistical methods were used to predetermine sample sizes. Sample sizes were determined by the number of biological replicates that are necessary for ensuring statistical significance. To minimize experimenter bias and ensure technical reproducibility, electrophysiological and behavioral experiments were independently performed by at least two trained experimenters, with each experimental condition replicated a minimum of twice. All data were collected and assembled by instrument software without interference.

All statistical analyses were conducted using Prism v9.5 (Graphpad, San Diego, CA). Two-tailed Student's unpaired t test was employed for comparisons between two distinct samples, while the paired t test was utilized for data involving repeated measurements from the same sample. Comparisons among three or more distinct samples were performed using one-way analysis of variance (ANOVA), followed by the Tukey post-hoc test. A significance threshold of $P < 0.05$ was applied. Data are presented as mean ± standard error of the mean (SEM) or as individual plots.

Ethics statements

Animal care and experiments were performed in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were approved by the Animal Care and Use Committee of Shanghai Medical College of Fudan University (mice) and the Institutional Animal Care and Use Committee (IACUC) of InnoStar Bio-tech Co., Ltd (monkey). The ethics approval number for mice experiments is 20211028-016. The ethics approval number for monkey experiments is IACUC-2024-M-791a and IACUC-2024-M-996.

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SUPPLEMENTARY MATERIALS

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Supplementary Text; Figs. S1 to S34; Tables S1 to S4; Reference (78); MDAR Reproducibility Checklist; Movies S1 to S4

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Tellurium nanowire retinal nanoprosthesis improves vision in models of blindness

Shuiyuan Wang, Chengyong Jiang, Yiye Yu, Zhenhan Zhang, Ruge Quhe, Ruyi Yang, Yufei Tian, Xindong Chen, Wenqiang Fan, Yinge Niu, Biao Yan, Chunhui Jiang, Yang Wang, Zhen Wang, Chunsen Liu, Weida Hu, Jiayi Zhang, and Peng Zhou

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Editor's summary

Prosthetic retinal implants hold promise for improving vision in blind and visually impaired individuals. Wang *et al*. developed a neuroprosthesis using tellurium nanowires (TeNWNs) that allowed efficient photovoltaic conversion in both the visible and the infrared range (see the Perspective by Fernández). TeNWNs showed good biocompatibility and did not require external power sources. The neuroprosthesis was able to improve vision in mouse and nonhuman primate models. The use of TeNWNs could substantially improve the efficacy of retinal neuroprostheses in restoring vision in blind and visually impaired individuals. —Mattia Maroso

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